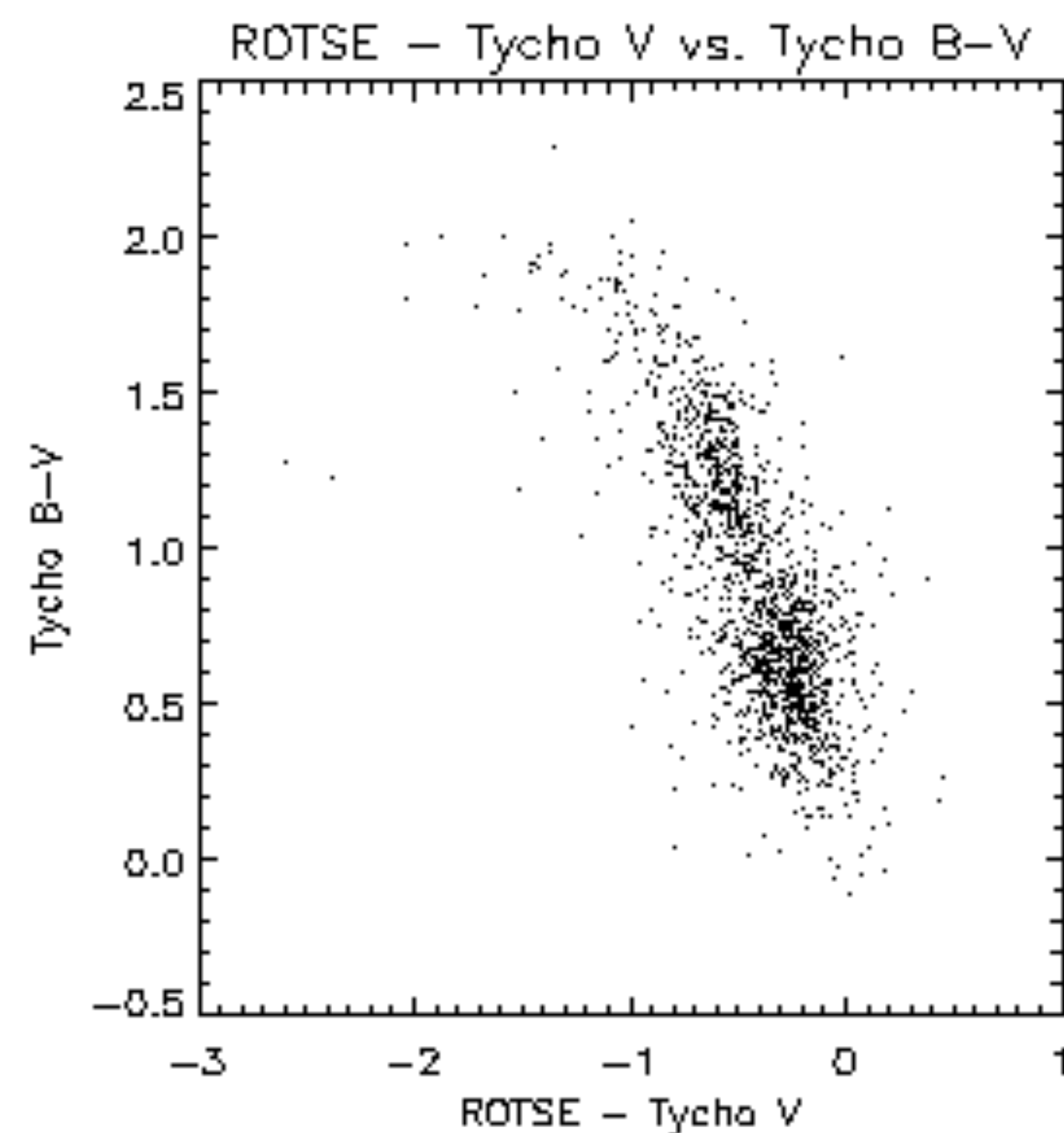
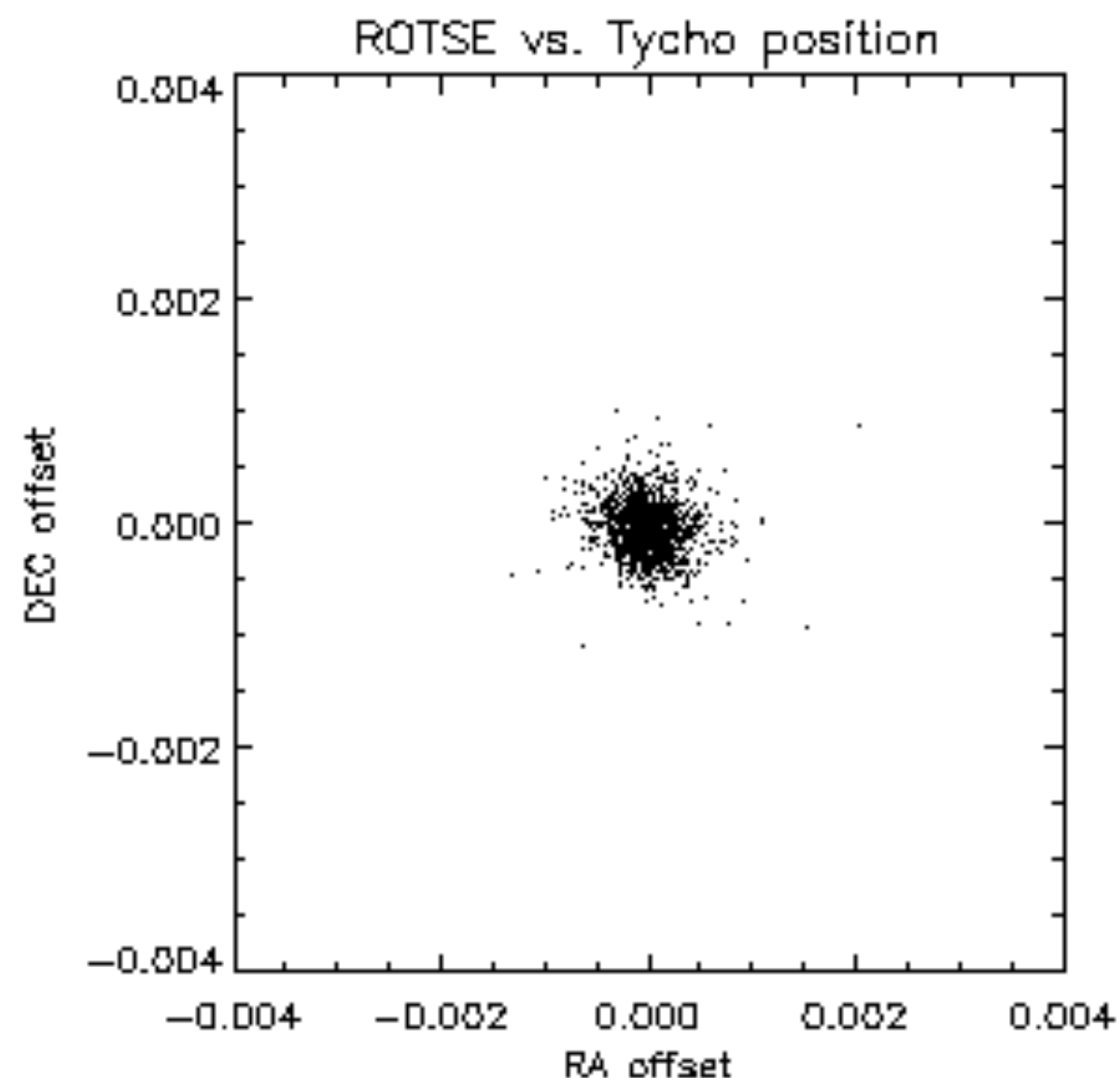




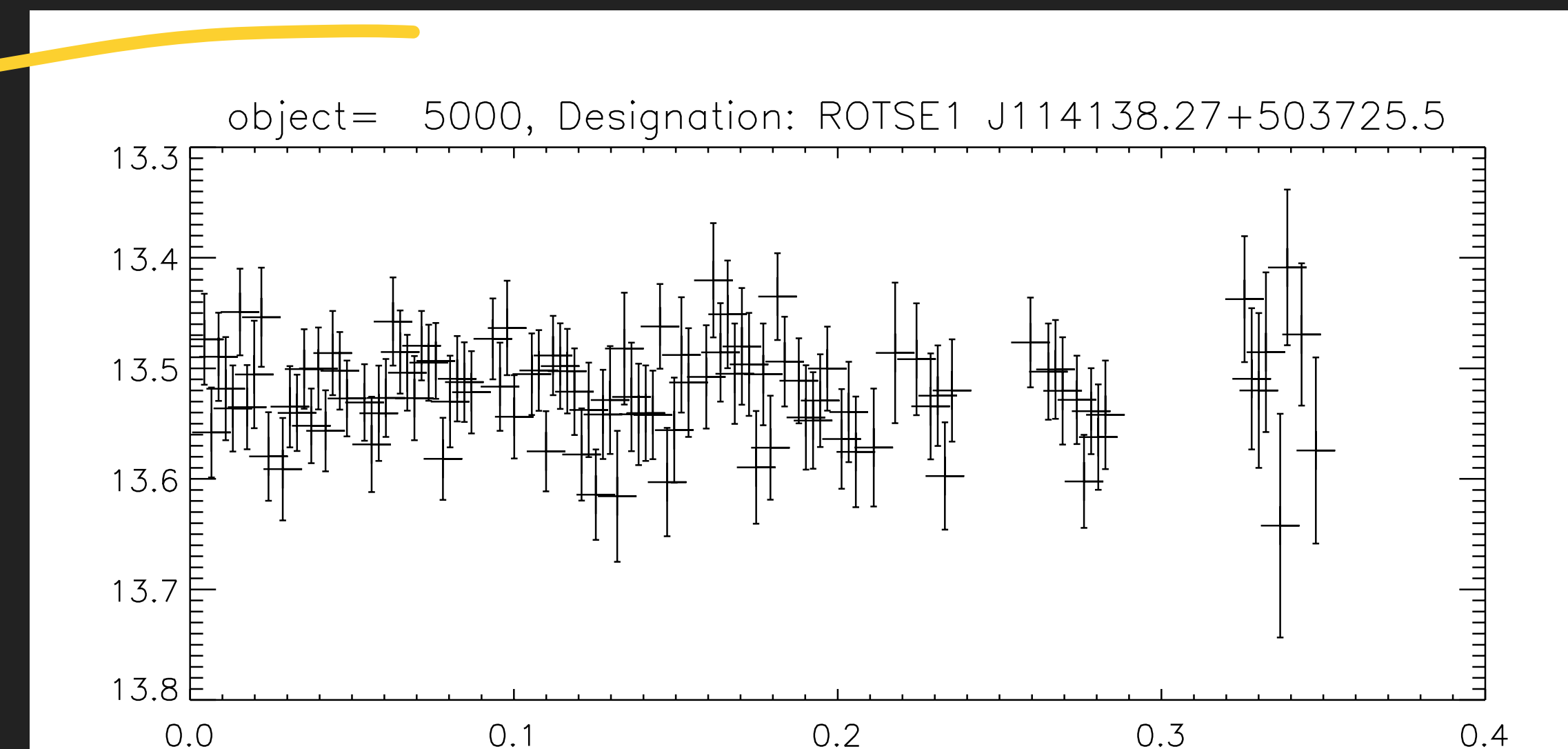
- ▶ IDL Basics
 - ▶ Selecting candidate lightcurves
 - ▶ Statistical concepts
 - ▶ ID OBJECTs of interest
- ▶ SN photometry
- ▶ Telescope performance
 - ▶ CCDs, astrometry, photometry

LAB 1

LIGHTCURVES

LIGHTCURVES

- ▶ Now select an object number – it doesn't matter which one for now. I'll pick 5000.
- ▶ Run a procedure 'find_burst' to extract the data and plot a lightcurve for this object
- ▶ IDL> `x=find_burst(match, 0,0,objid=5000,log='firsttry')`
- ▶ IDL> `$ls *.ps`
- ▶ IDL> `$evince firsttry.ps &`
- ▶ Let's have a look at your plot!



LAB 2: FINDING VARIABLE STAR CANDIDATES

SOURCE SELECTION

- ▶ Picking up where we left off, let's examine our match structure further to select lightcurves that appear to be varying

- ▶ We use 'find_burst', but this time we will use it to select objects that are statistically different than constant

- ▶ 3 statistical quantities we can cut on to select variable candidates

- ▶ **delta_magnitude**: the difference in magnitude between the brightest and dimmest measurement in a lightcurve: $\Delta m = m_{max} - m_{min}$

- ▶ **Maximum significance**: the significance of the maximum variation: $\sigma_{max} = \frac{(m_{max} - m_{min})}{\sqrt{\epsilon_{max}^2 + \epsilon_{min}^2}}$

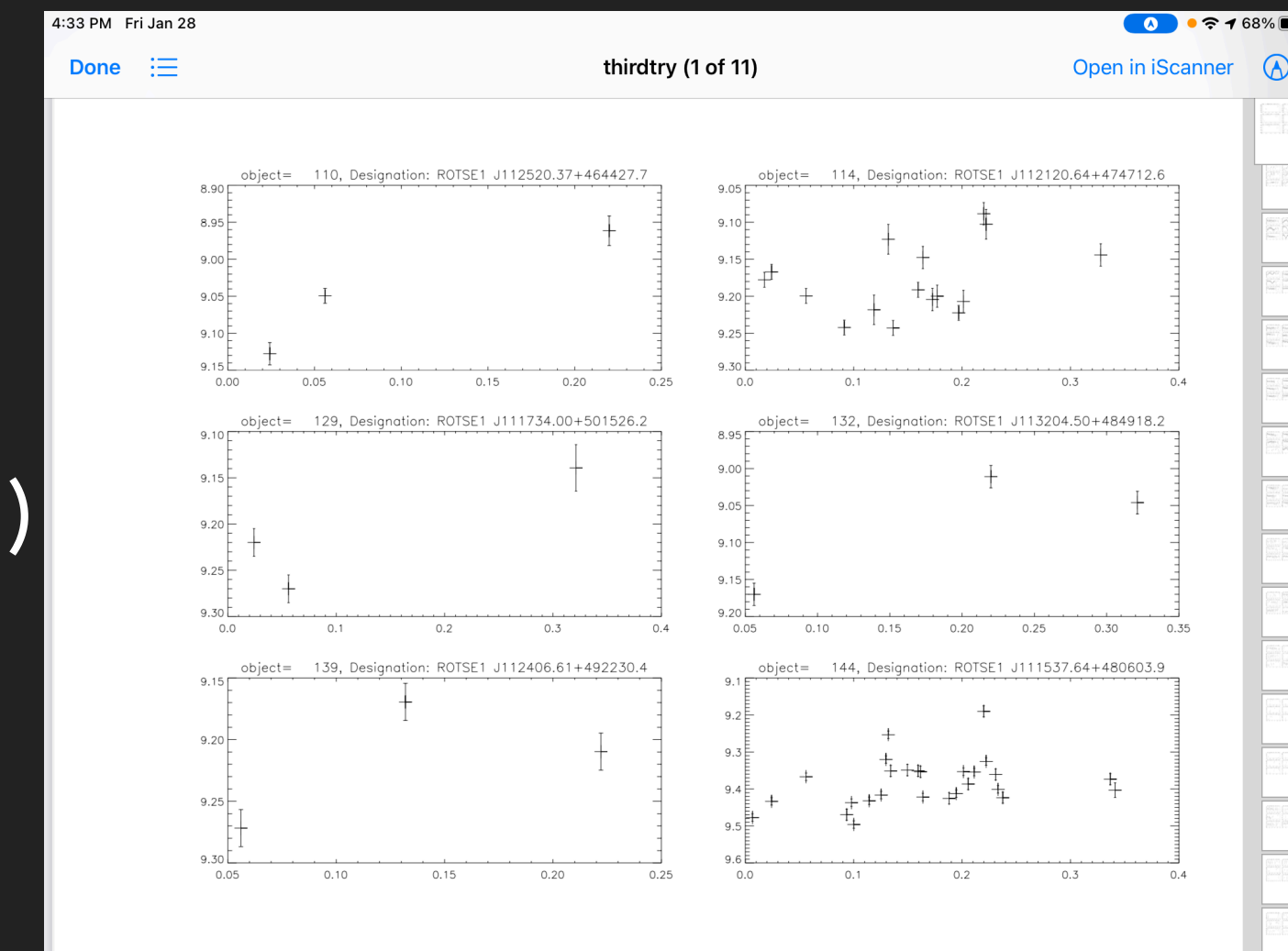
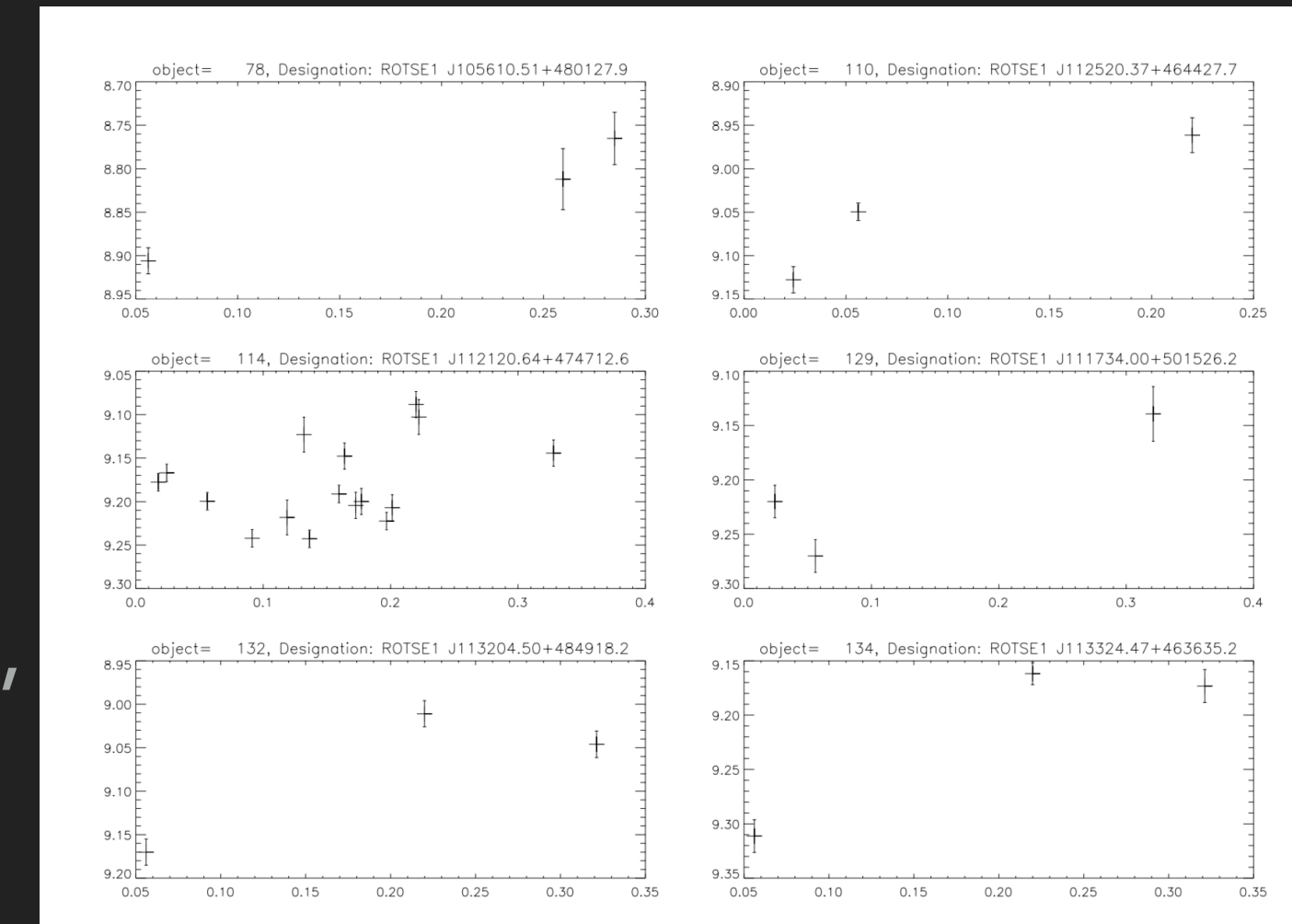
- ▶ **Minimum chi-squared**: the square of the difference of each measurement from the median of a lightcurve, divided by the square of the statistical uncertainty on this difference:

$$\chi^2 = \sum \left[\frac{(m_i - m_{avg})}{\sigma_i} \right]^2$$

- ▶ Values are summed up over all measurements in a lightcurve and divided by the number of measurements minus 2, to give a 'χ2 per degree of freedom'
- ▶ You can find more about these in the VSP paper: E. Fagg, et al., Journal of Undergraduate Research in Physics (2009).

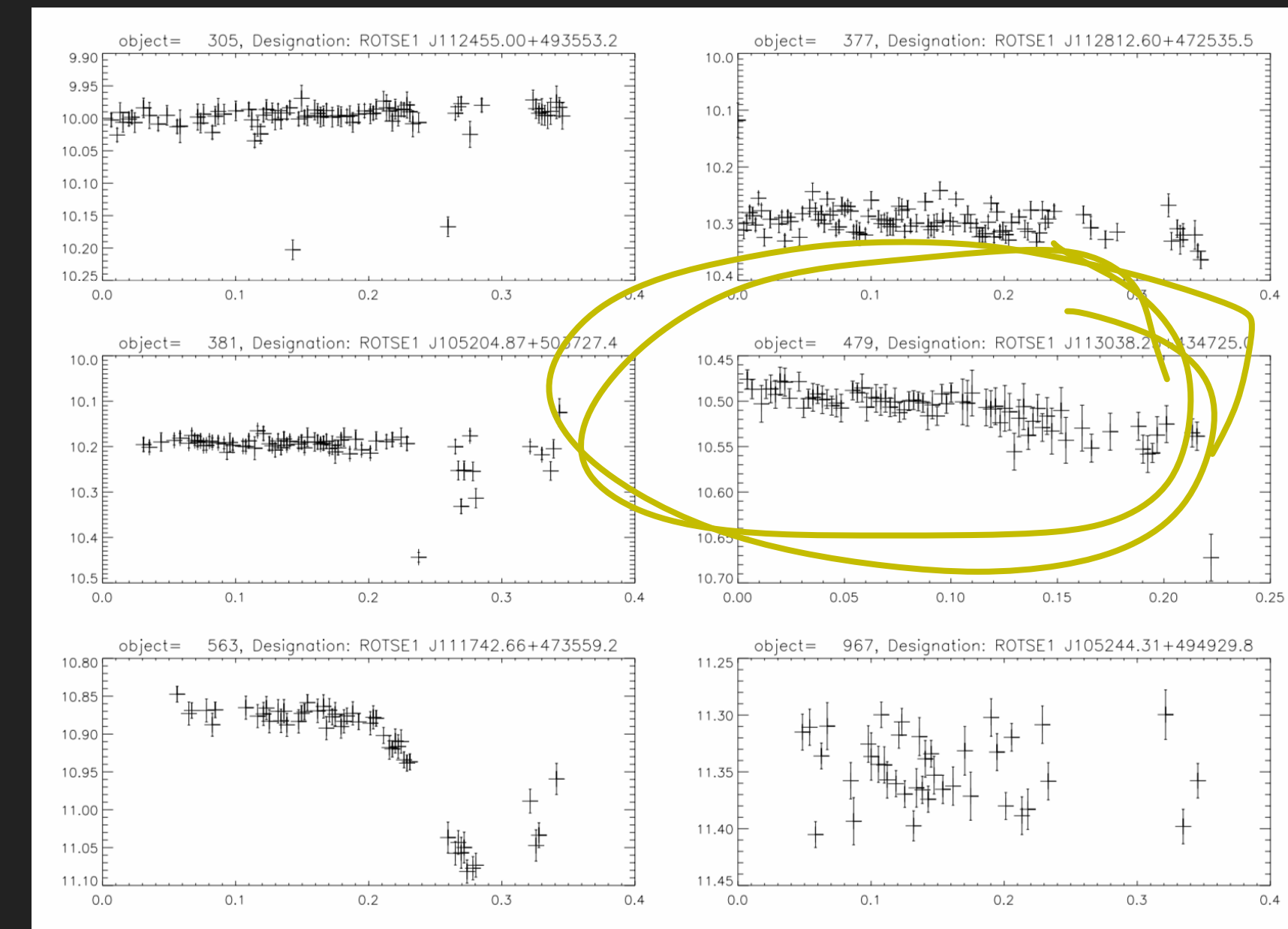
FINDING CANDIDATE VARIABLES

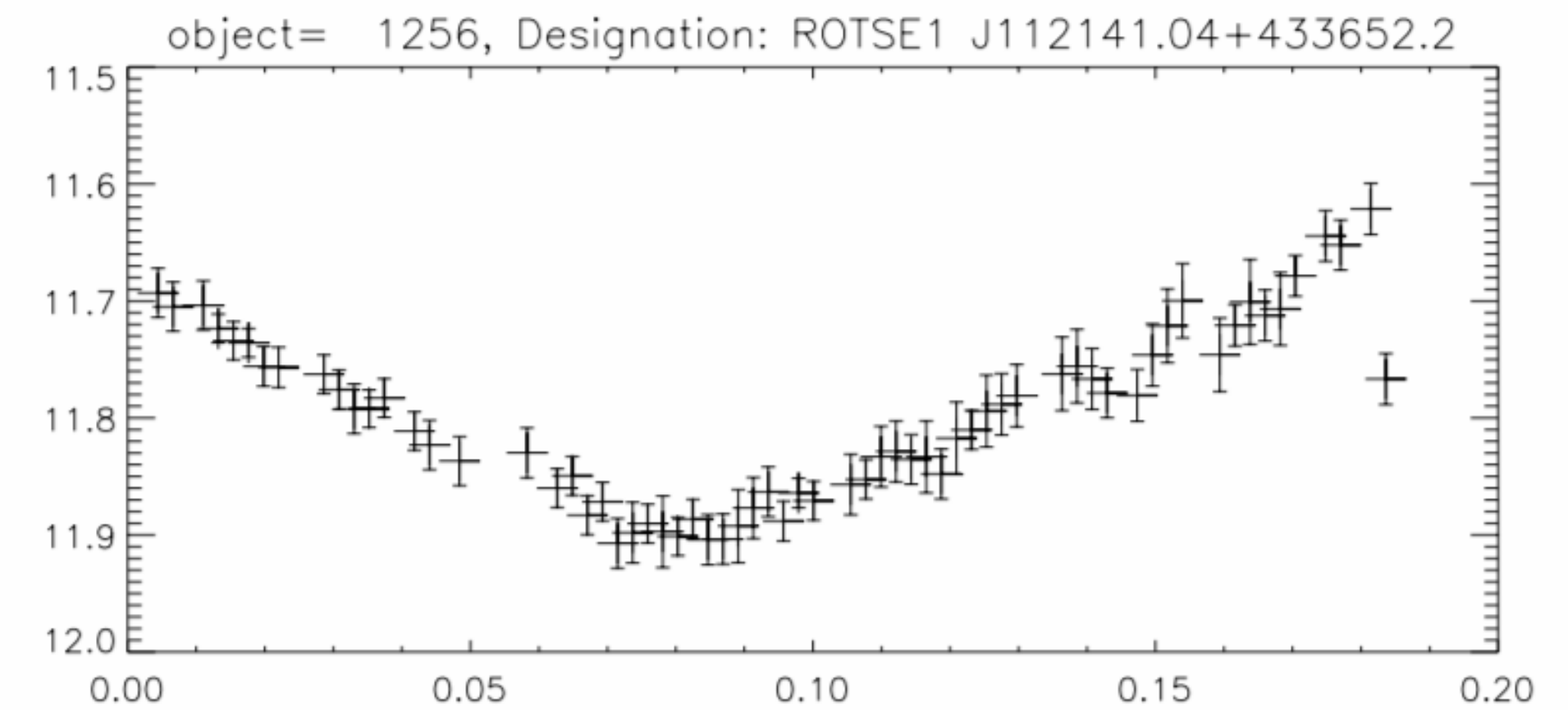
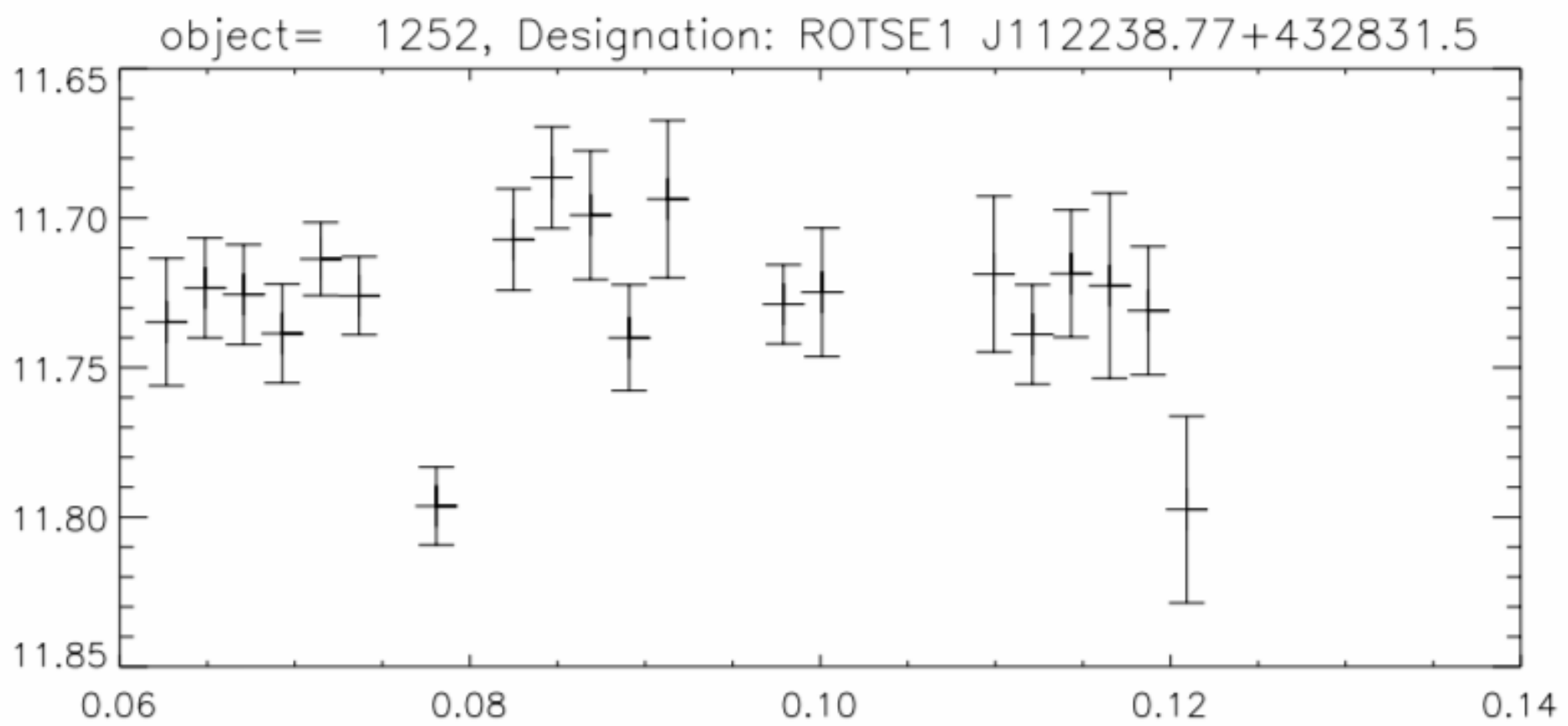
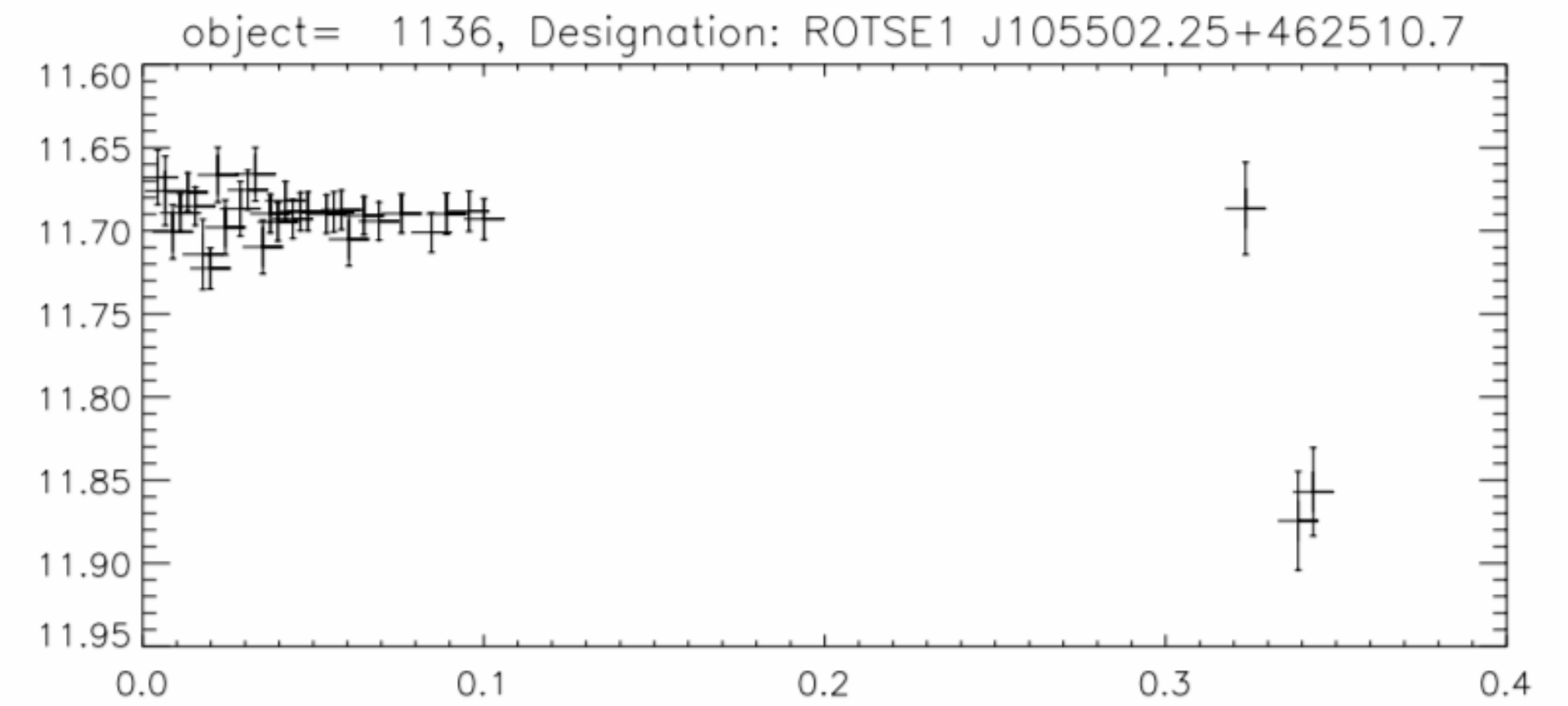
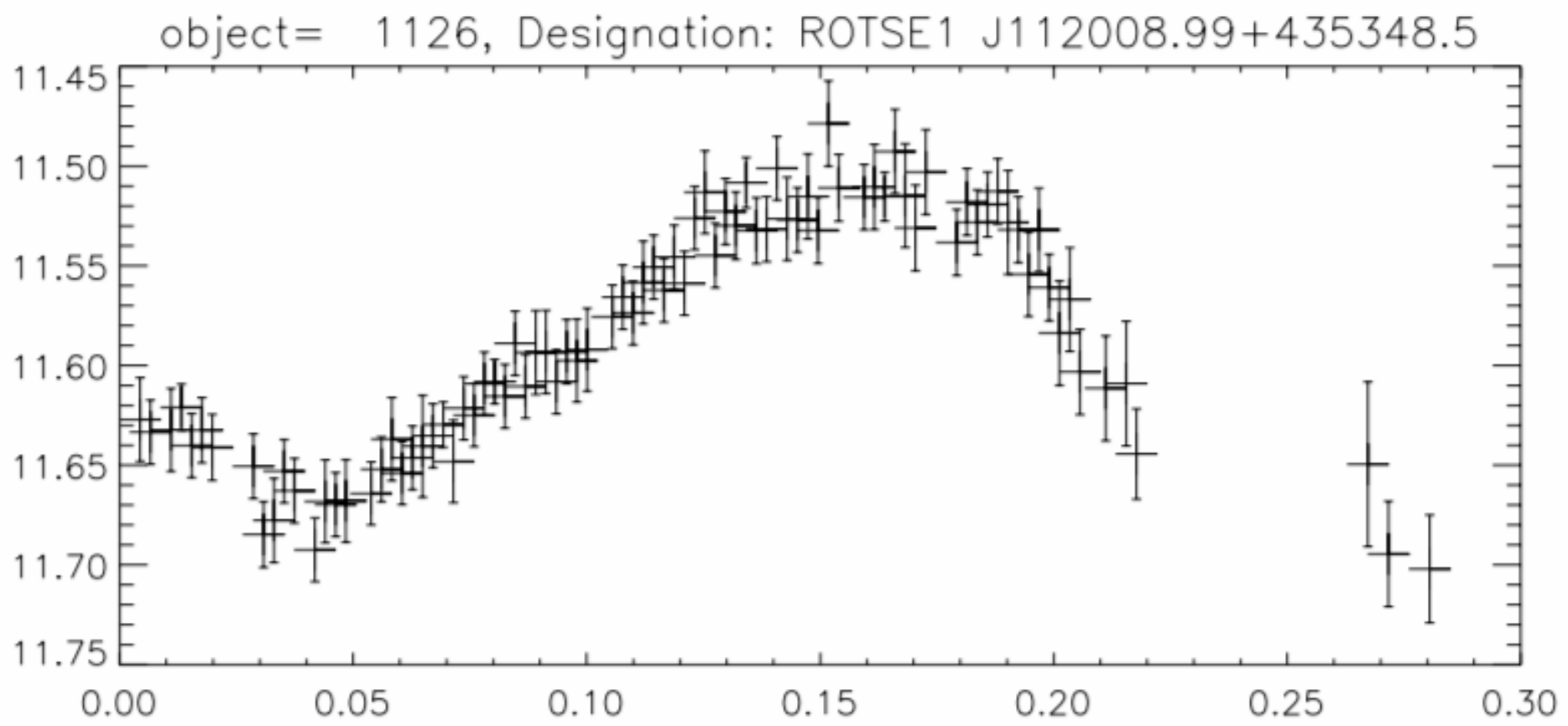
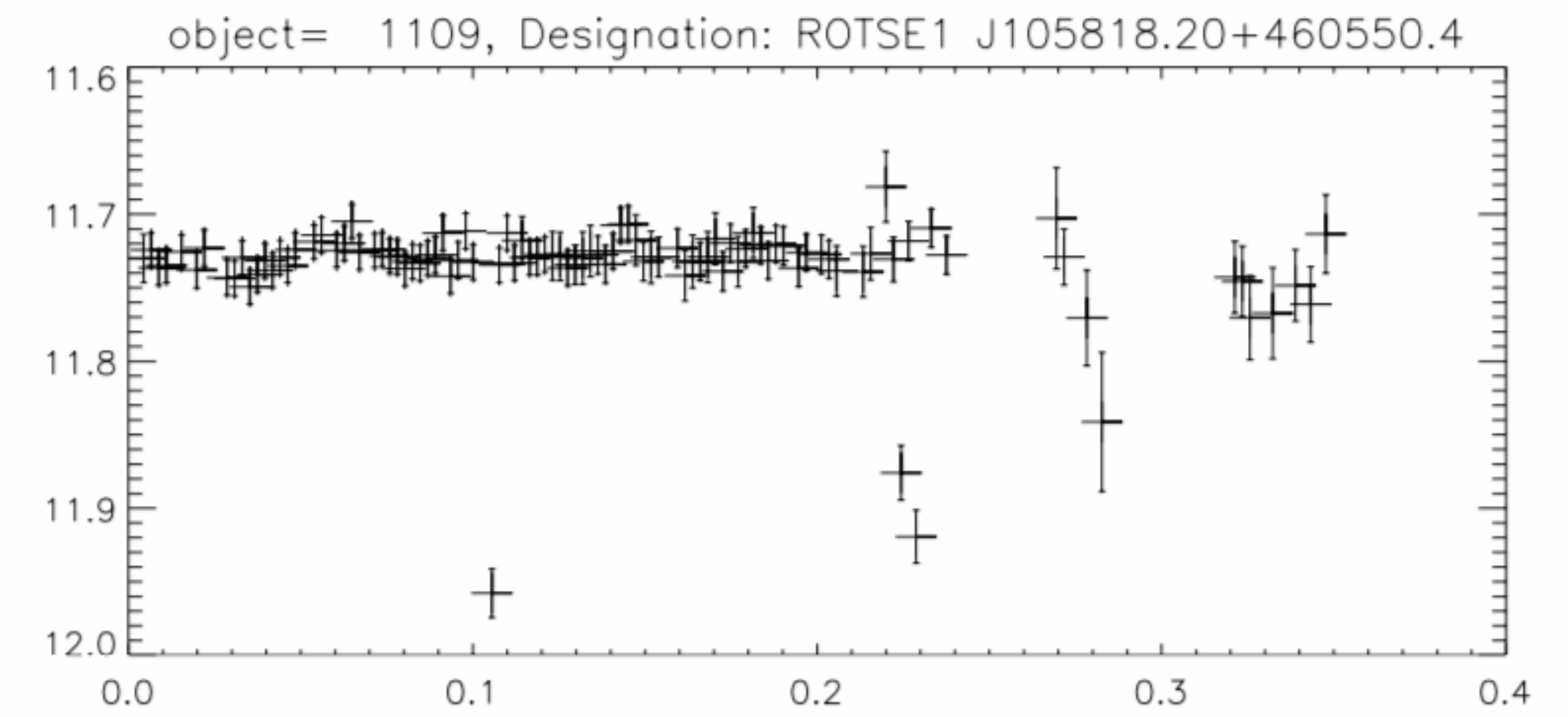
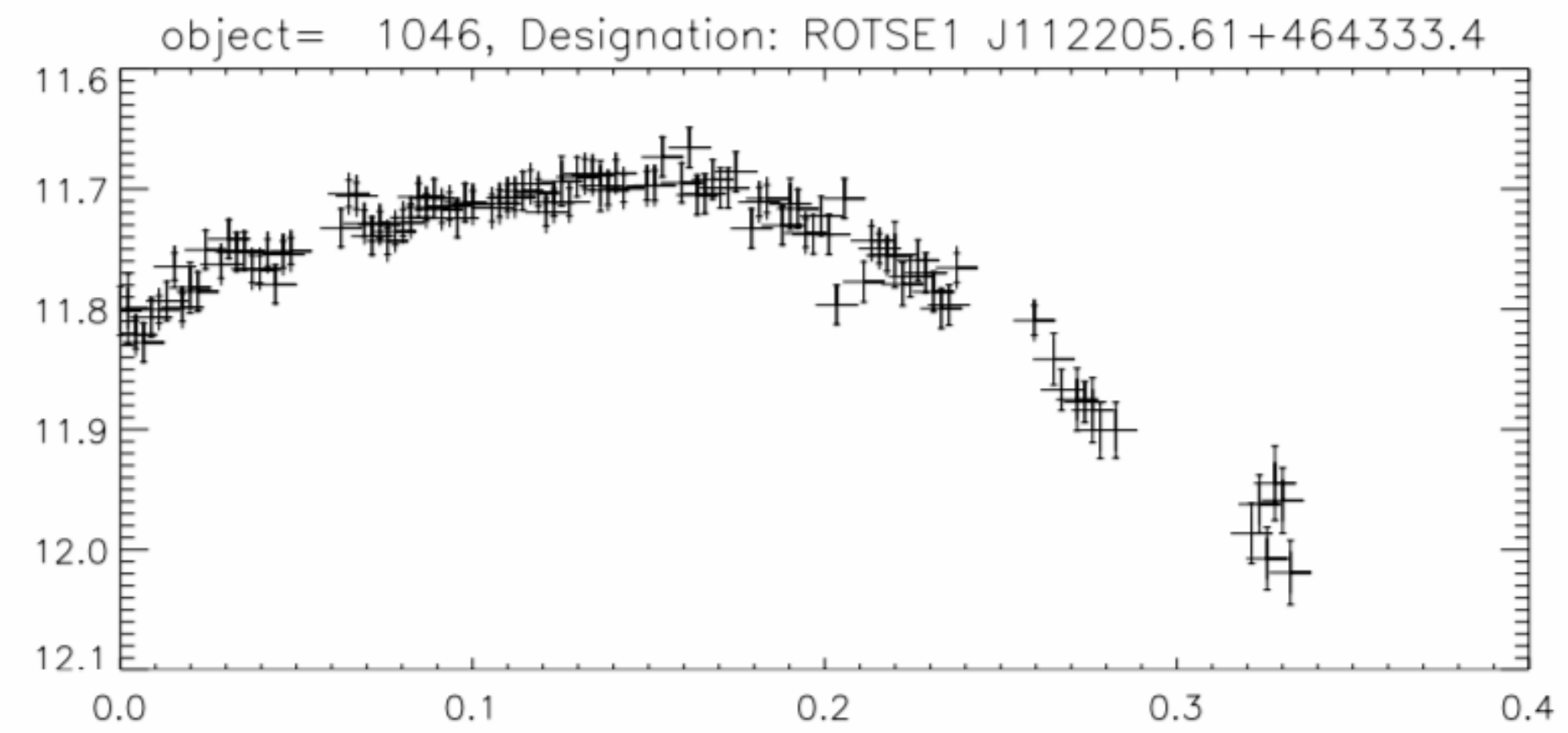
- ▶ How do we use find_burst in this way?
 - ▶ IDL> x=find_burst(match, 0.1,3.0,log='secondtry')
 - ▶ Where 2nd and 3rd arguments are delta_mag and max_sig, respectively
 - ▶ Examine outputs to see what you get:
 - ▶ IDL> \$evince secondtry.ps &
 - ▶ Examine this file, in all it's pages.
 - ▶ How many objects are found?
 - ▶ What happens when you vary delta_mag? Max_sig?
 - ▶ Now let's add a powerful variable, chi-squared:
 - ▶ IDL> x=find_burst(match, 0.1,3.0,minchisq=2.0,log='thirdtry')
 - ▶ What happens to the # of candidates now?



FINDING CANDIDATE VARIABLES

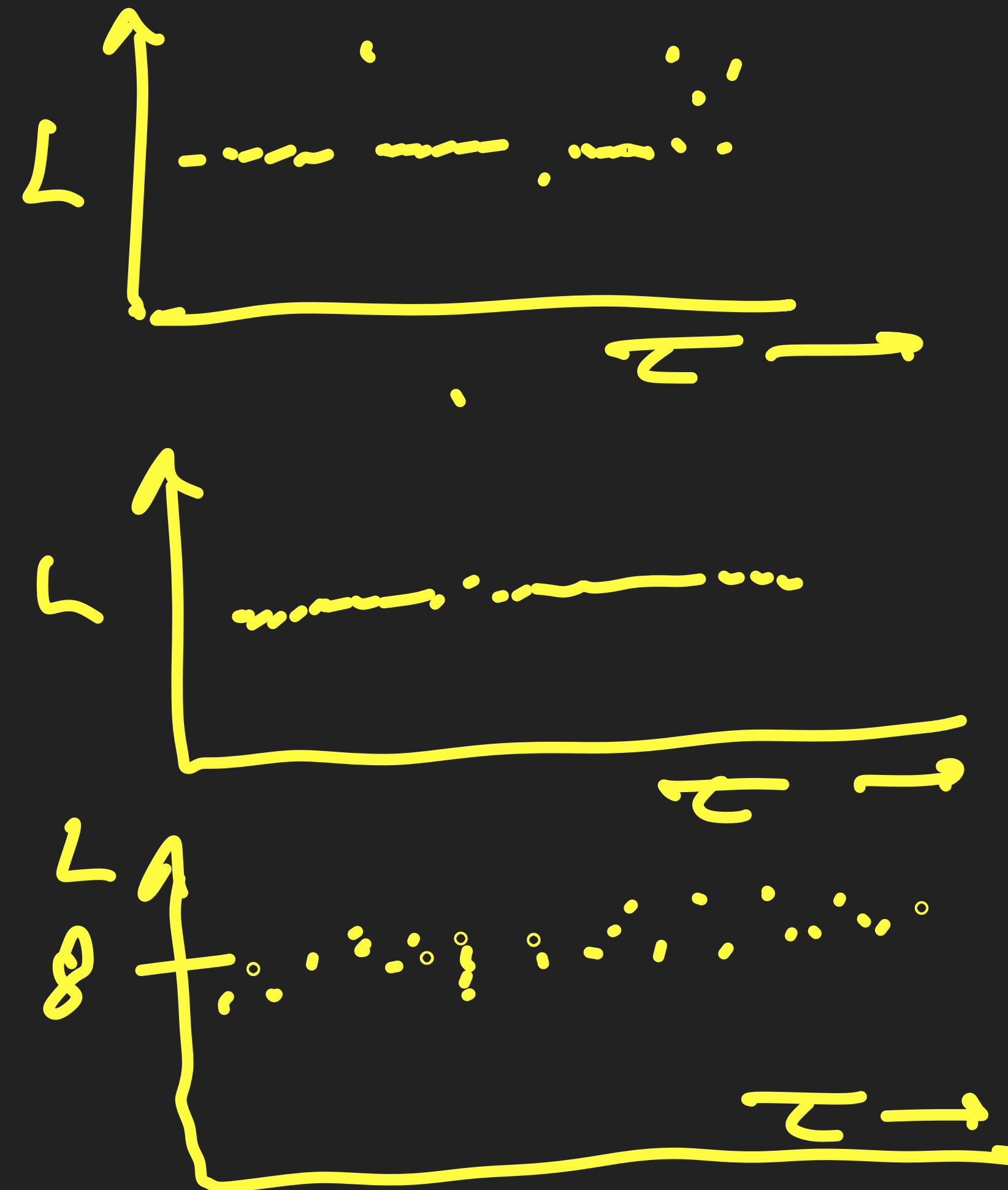
- ▶ Now, look at the individual lightcurves for all pages in your .ps files
- ▶ Can you pick out lightcurves that are physical, and those that may be unphysical?
 - ▶ Why are there lightcurves that are not physical?
 - ▶ What is happening there?
 - ▶ What physics could produce the physical lightcurves?
- ▶ You will notice a lot of 'background' from apparent variations that have nothing to do with variables
 - ▶ How do we handle these?
 - ▶ We will revisit this next time.





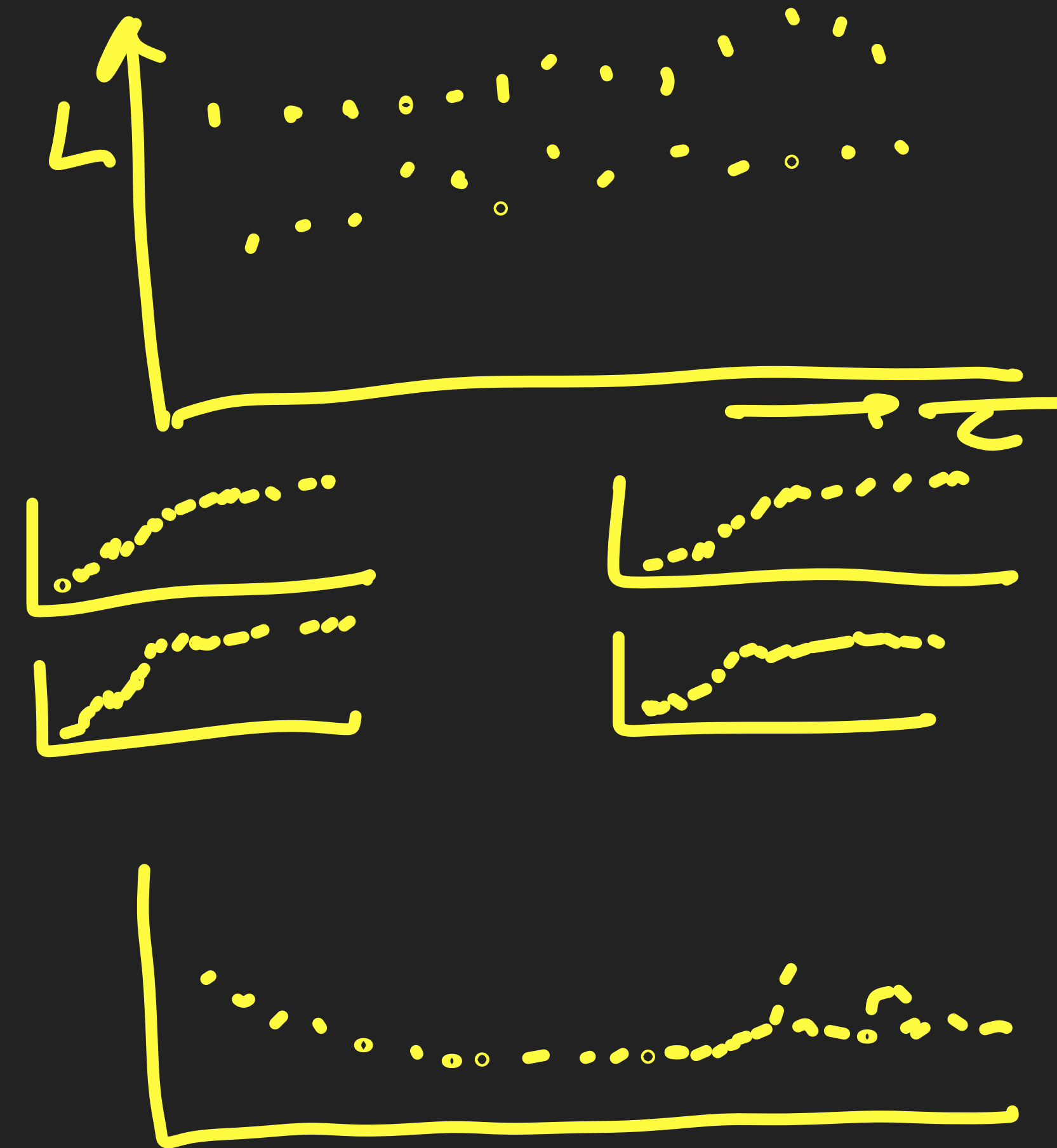
INSTRUMENTAL CANDIDATES IN LIGHTCURVES

- ▶ **Hot pixel** - occasional, erratic change in brightness in lone exposures
- ▶ **Cosmic rays** - single brightened observation
- ▶ **Saturation** - all bright stars seem variable (mag <9-10)



INSTRUMENTAL CANDIDATES IN LIGHTCURVES (CONT)

- ▶ **Blending** - alternating magnitudes statistically DES repaint from each other
- ▶ **Weather** vs. time - slow trend over night, but for several nearby objects in a camera field
- ▶ **Vignetting** vs. time - possibly slow trend or intermittent ('sticking') behavior in corners of camera field



HAND SCANNING

- ▶ Extracting a list of candidates for each night of a field requires us to further examine the lightcurves passing initial selection
 - ▶ A stars size helps us here: physical processes take time to communicate across such a large object
- ▶ Confirmation: require each detection to be contiguous in time with another detection.
- ▶ Consistency: require consecutive exposures to be statistically compatible with each other
- ▶ Continuity: apparent variations should manifest as a series of observations with an overall trend. Generally, we look for these trends to be $O(10)$ minutes or longer



SUPPORTING MATERIAL

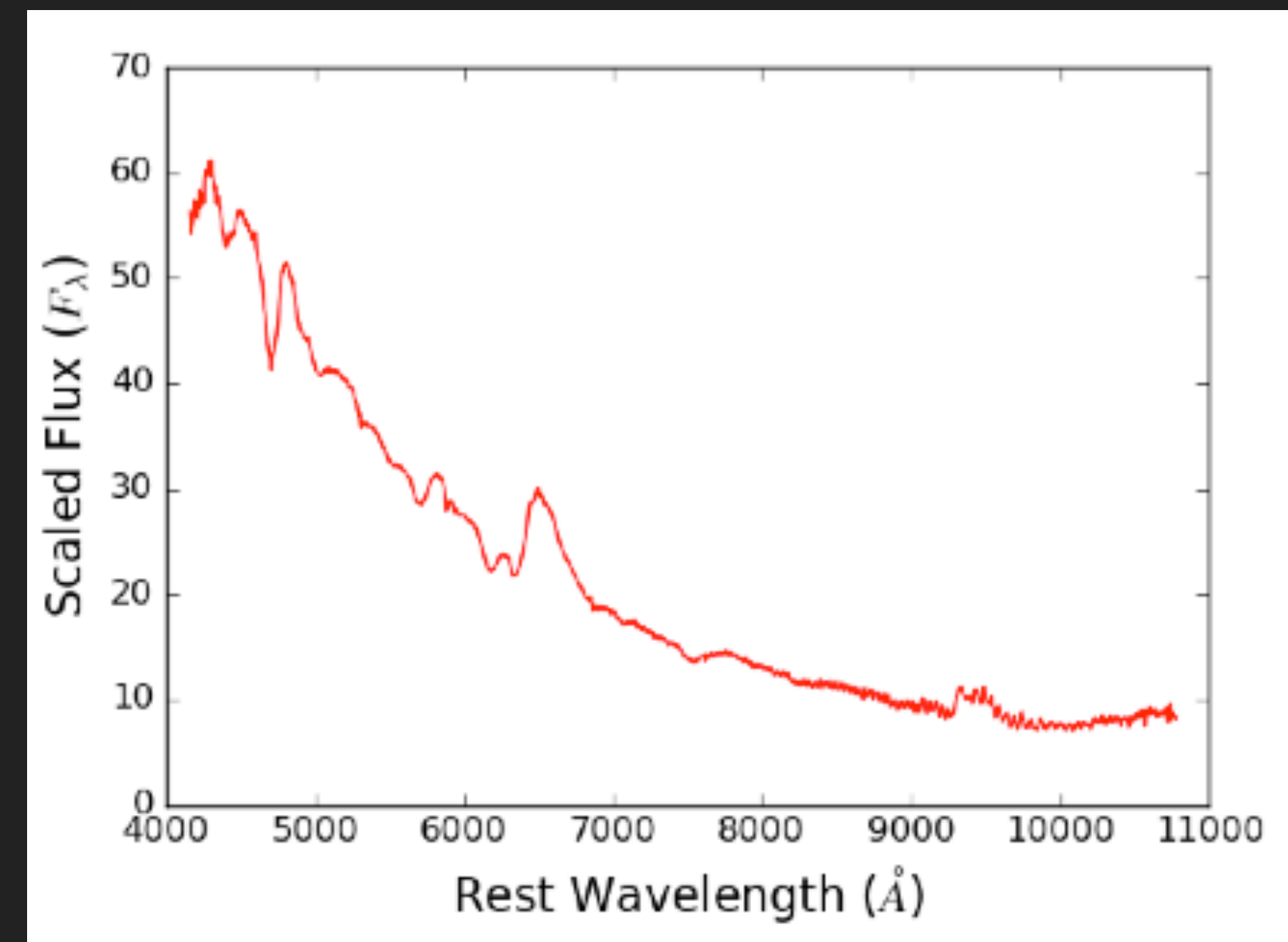
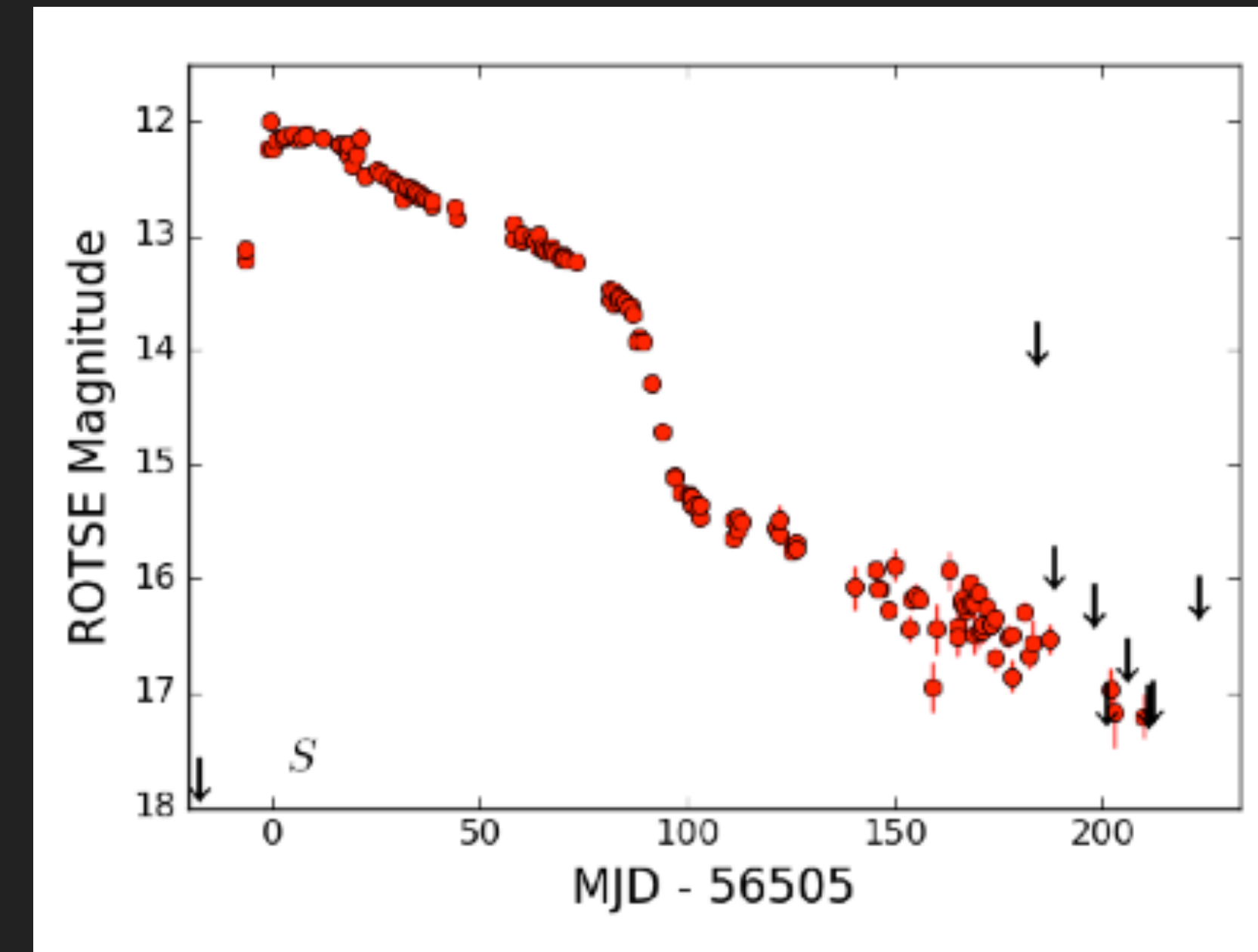
- ▶ There is a 'manual' written by former students using this data for research:
 - ▶ https://docs.google.com/document/d/1vEw-jgaDF4lw98pGBkhn5T5LEG255ynzC_aWDrKADUY
 - ▶ The first 3 chapters are useful for this course
- ▶ Also the web-site for ROTSE, rotse.net, has several references that may be of use at:
 - ▶ <http://rotse.net/publications/publications.html>
 - ▶ In particular, Kehoe et al. 2002 (source of the data for this course).

SN PROJECT: PROCEDURES AND FILES

- ▶ Log into analysis workstation in OIR Lab
 - ▶ > ssh -Y smurotse@rotse14.physics.smu.edu
 - ▶ For this week, use 'rotse1.physics.smu.edu' though
 - ▶ > password: 'grb990123'
- ▶ Go to the directory where the data structures are:
 - ▶ > cd x371/<your user name>
 - ▶ > ls
 - ▶ There will be two folders: 'sn-phot' and 'sn-spec'
 - ▶ sn-phot contains one or more lightcurve ASCII files
 - ▶ sn-spec contains two or more spectra files

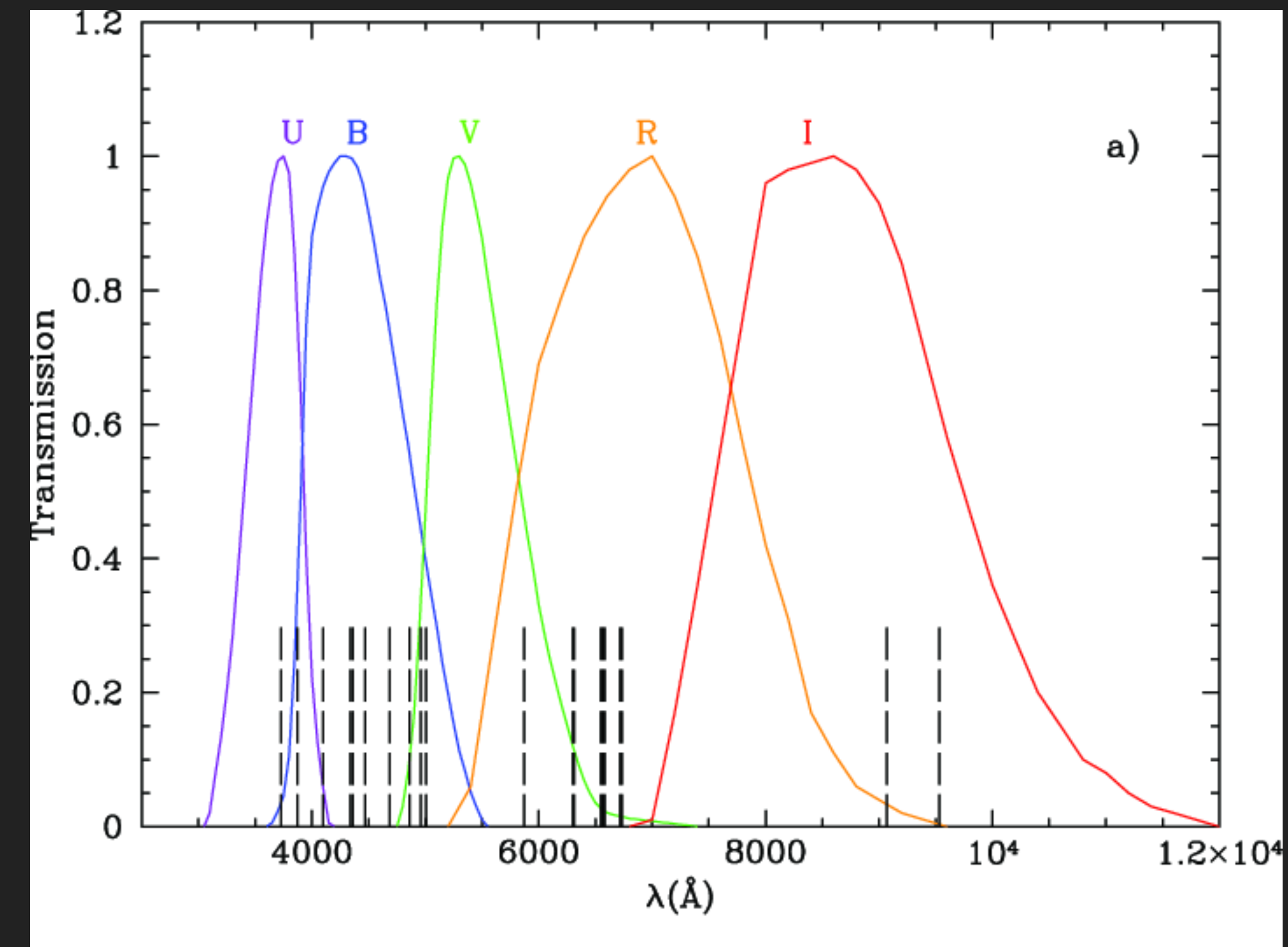
DATA TO EXAMINE

- ▶ New photometry file: *.dat
 - ▶ Columns for MJD, magnitude, error
 - ▶ Lightcurve from ROTSE-IIIb telescope
- ▶ Spectrum files named: *.flm
 - ▶ Spectra before, during and/or after peak
- ▶ Read in your data, and examine the contents
 - ▶ For this week, make a 1D plot of the lightcurve, and 1D plot of one of the spectra



BROADBAND MAGNITUDE AND TEMPERATURE

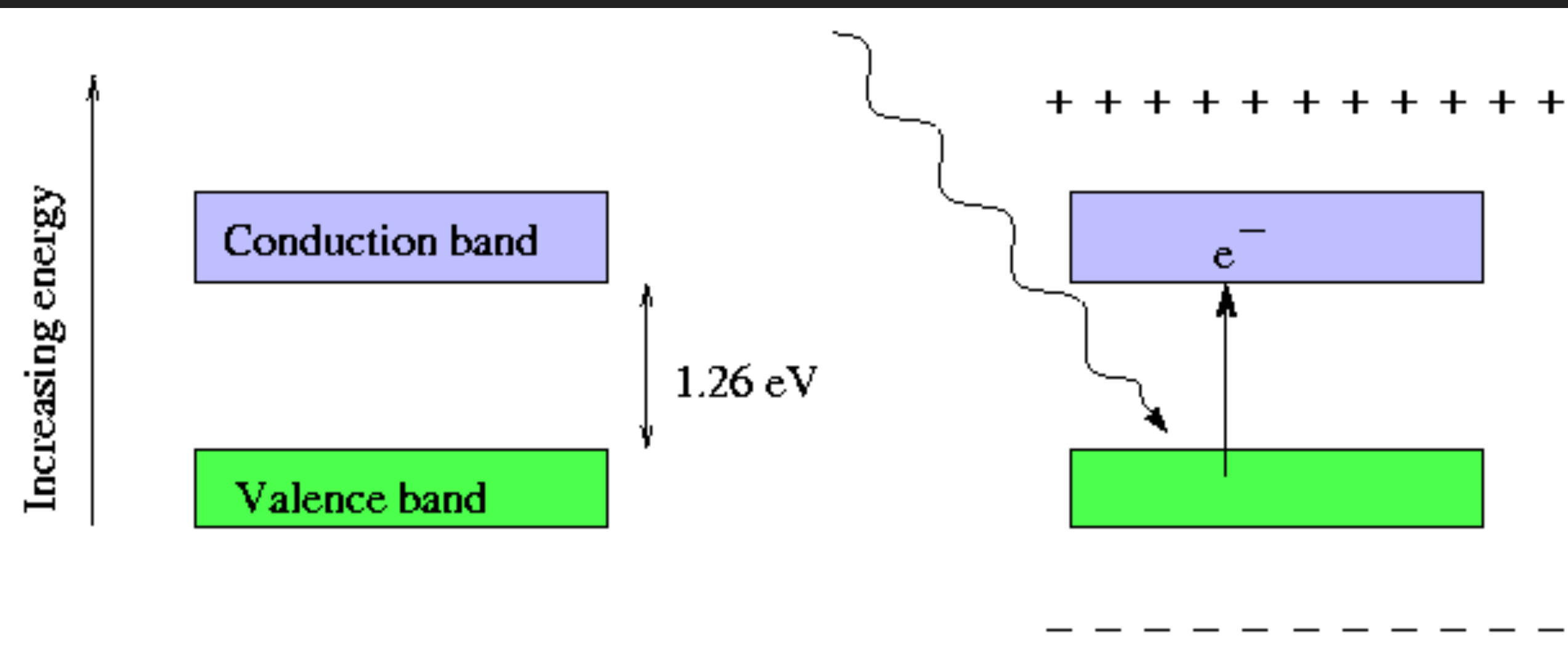
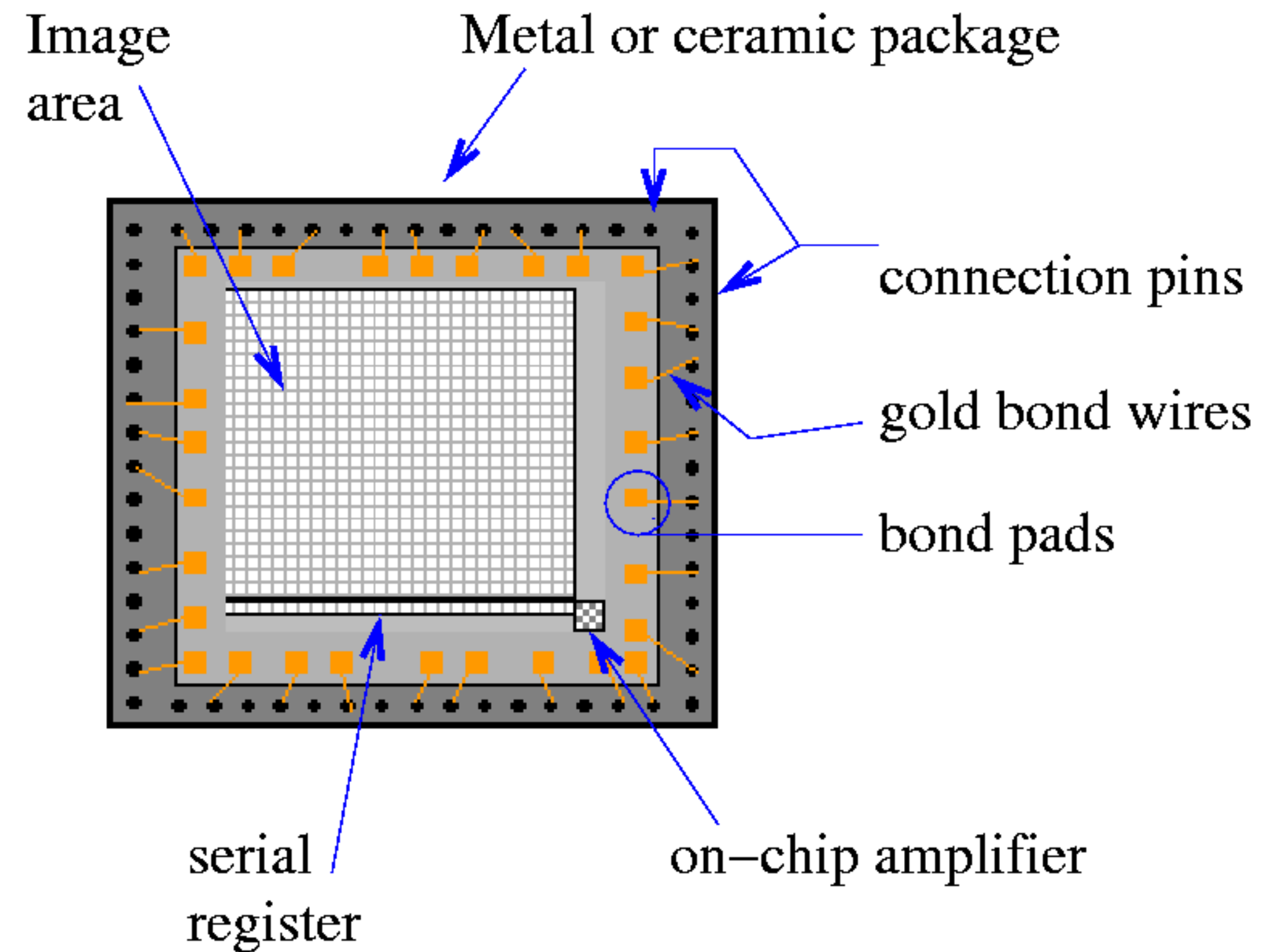
- ▶ Integrate spectrum over broadband
 - ▶ B passband
 - ▶ V passband
 - ▶ Need to convert from flux units/angstrom to magnitude
 - ▶ Do your V-band magnitudes agree roughly with the photometry?
- ▶ Calculate color index
 - ▶ B-V
 - ▶ At epoch you have spectrum



SUPPLEMENTARY MATERIAL

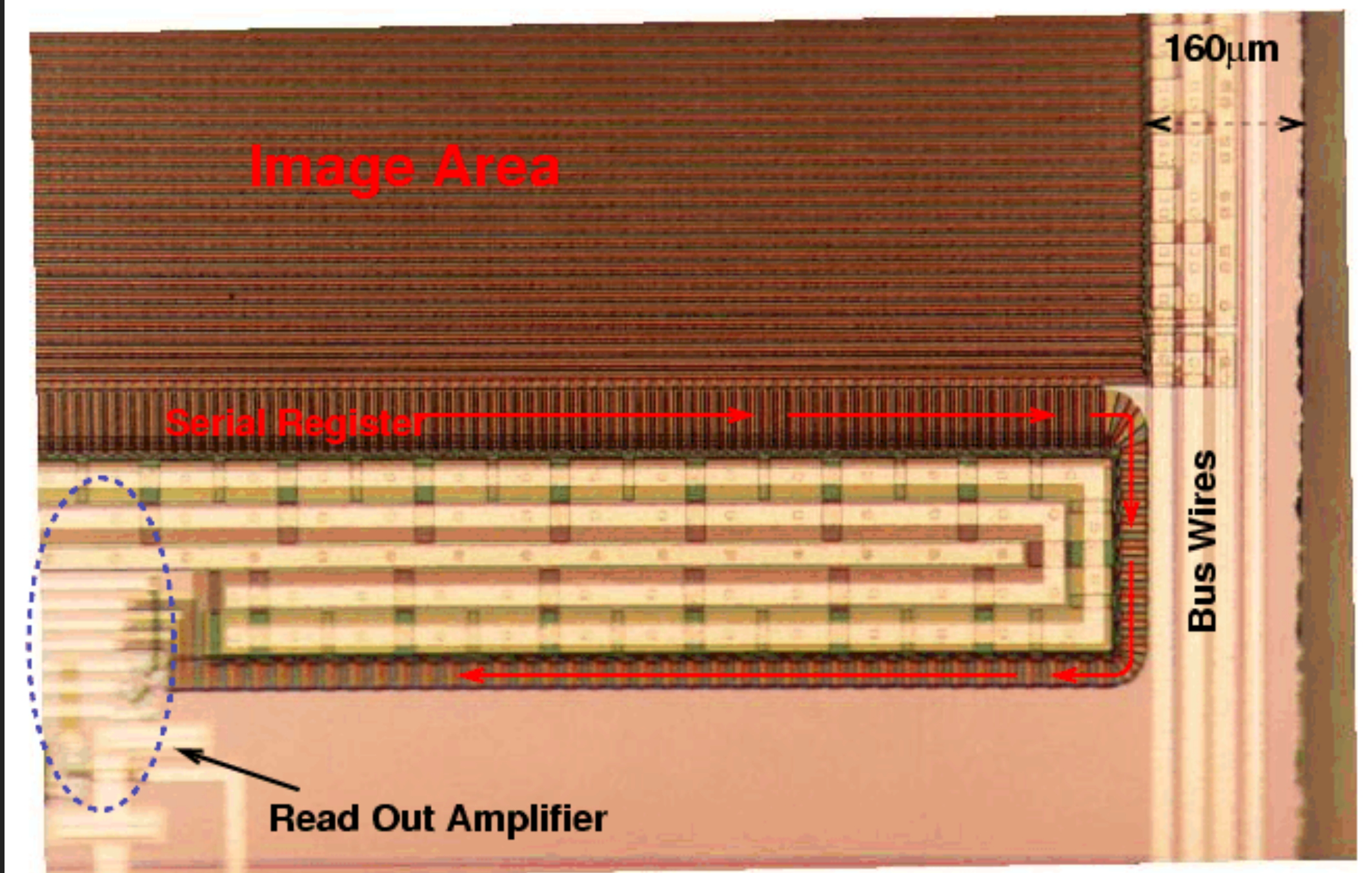
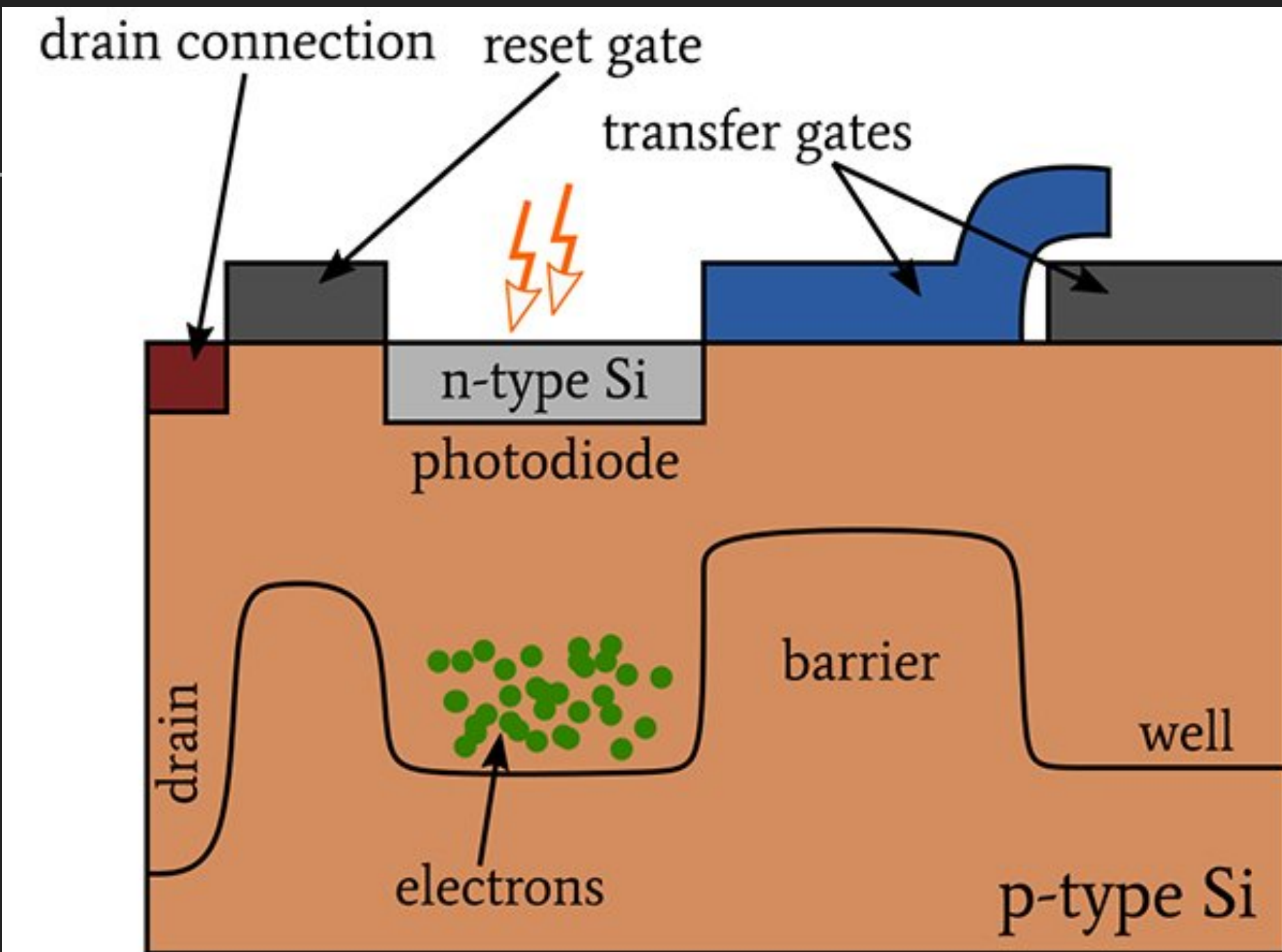
A CCD DETECTOR

- ▶ Charge-coupled device
- ▶ Imaging sensor
- ▶ 2D array of detector elements
 - ▶ p-n junction diodes
- ▶ Photons incident on p-type
 - ▶ Charge liberated
 - ▶ Moves electrons from valence to conduction band of p-type



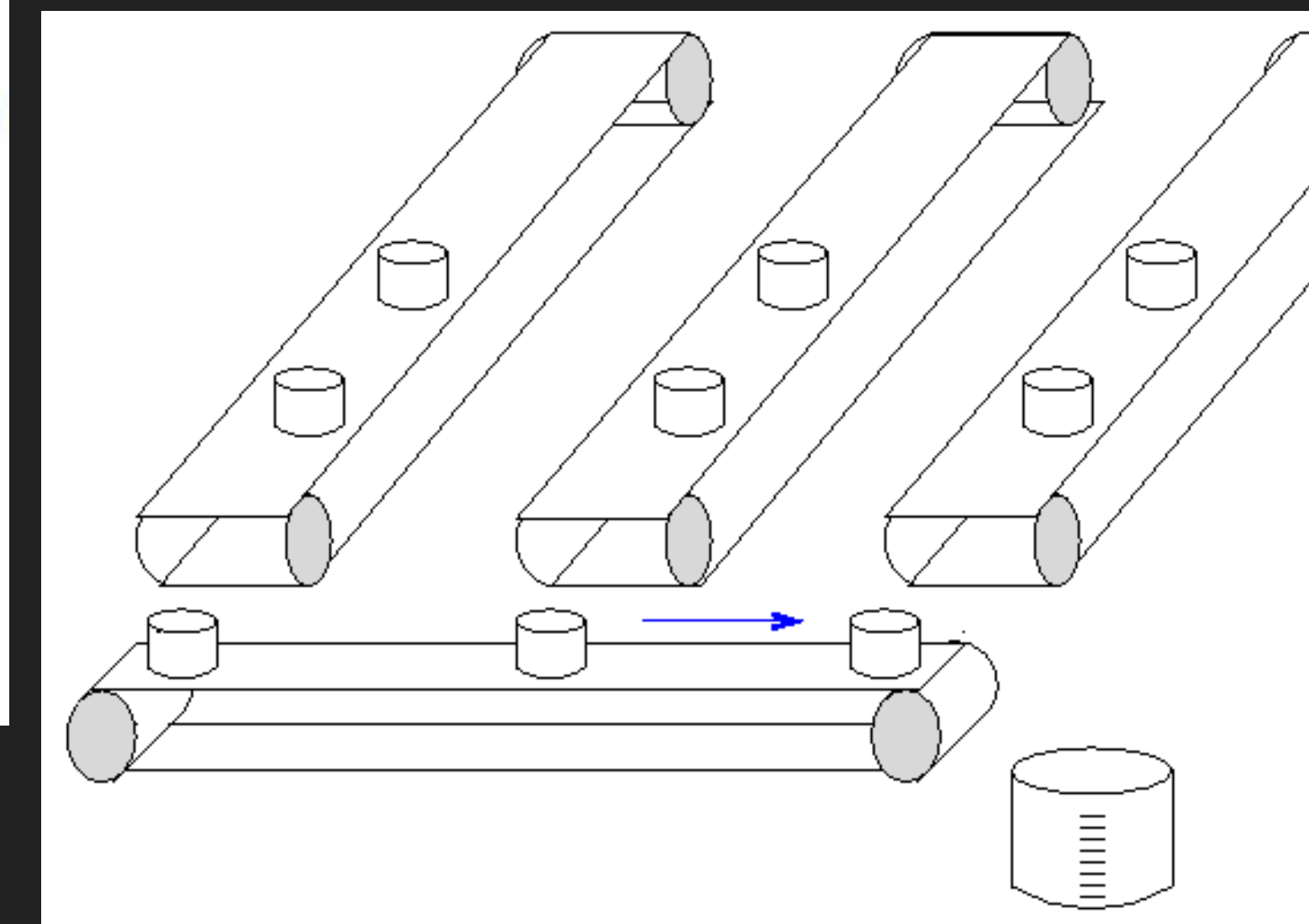
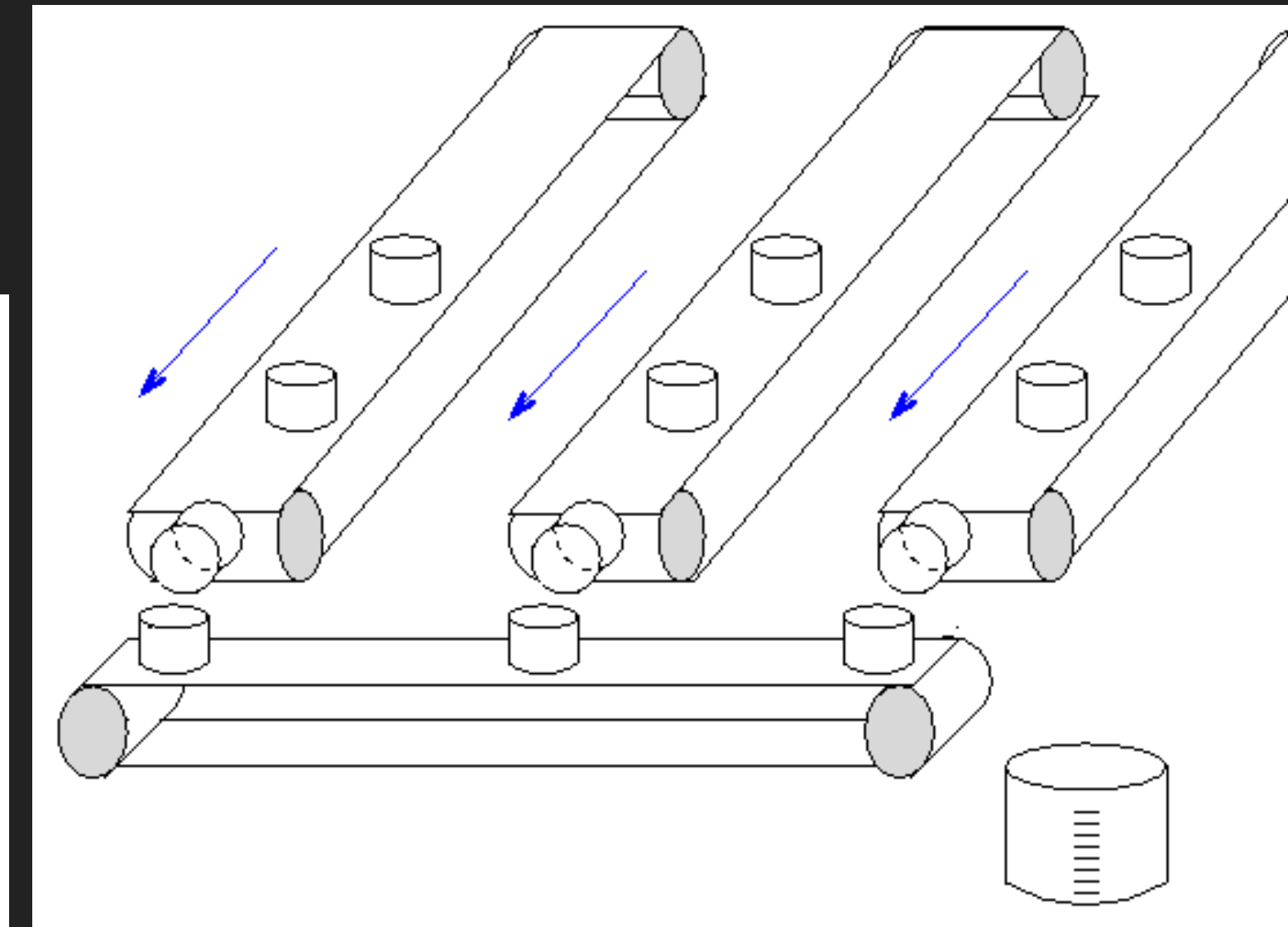
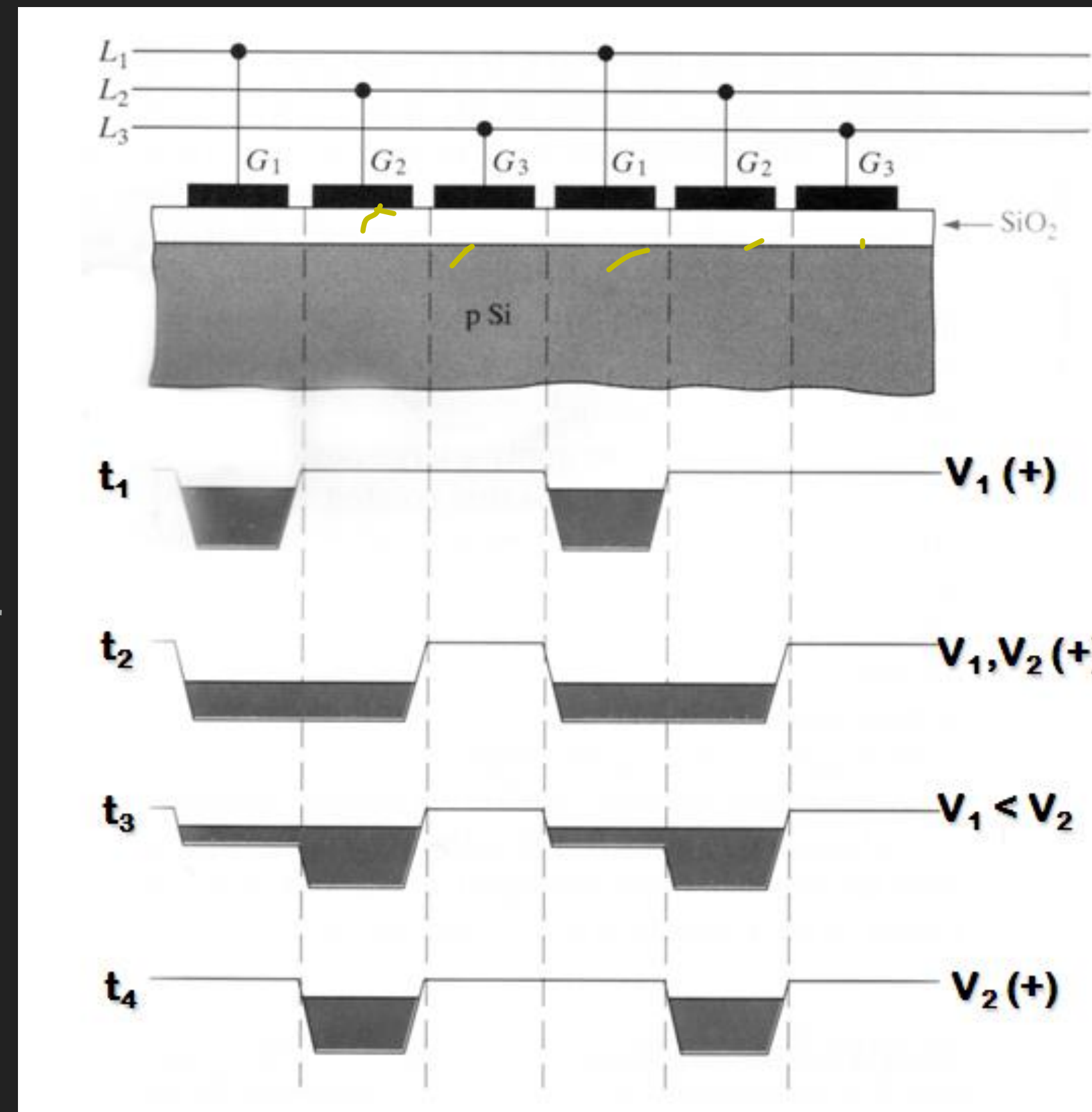
CCD OPERATION

- ▶ Structured so that voltages can be controlled
 - ▶ Columns
 - ▶ Rows
 - ▶ Prevent charge from moving from region it was liberated
 - ▶ A **pixel** like a 2D potential well holding charge
- ▶ At bottom of CCD
 - ▶ Readout register: not for imaging
 - ▶ Takes charge from row just above it



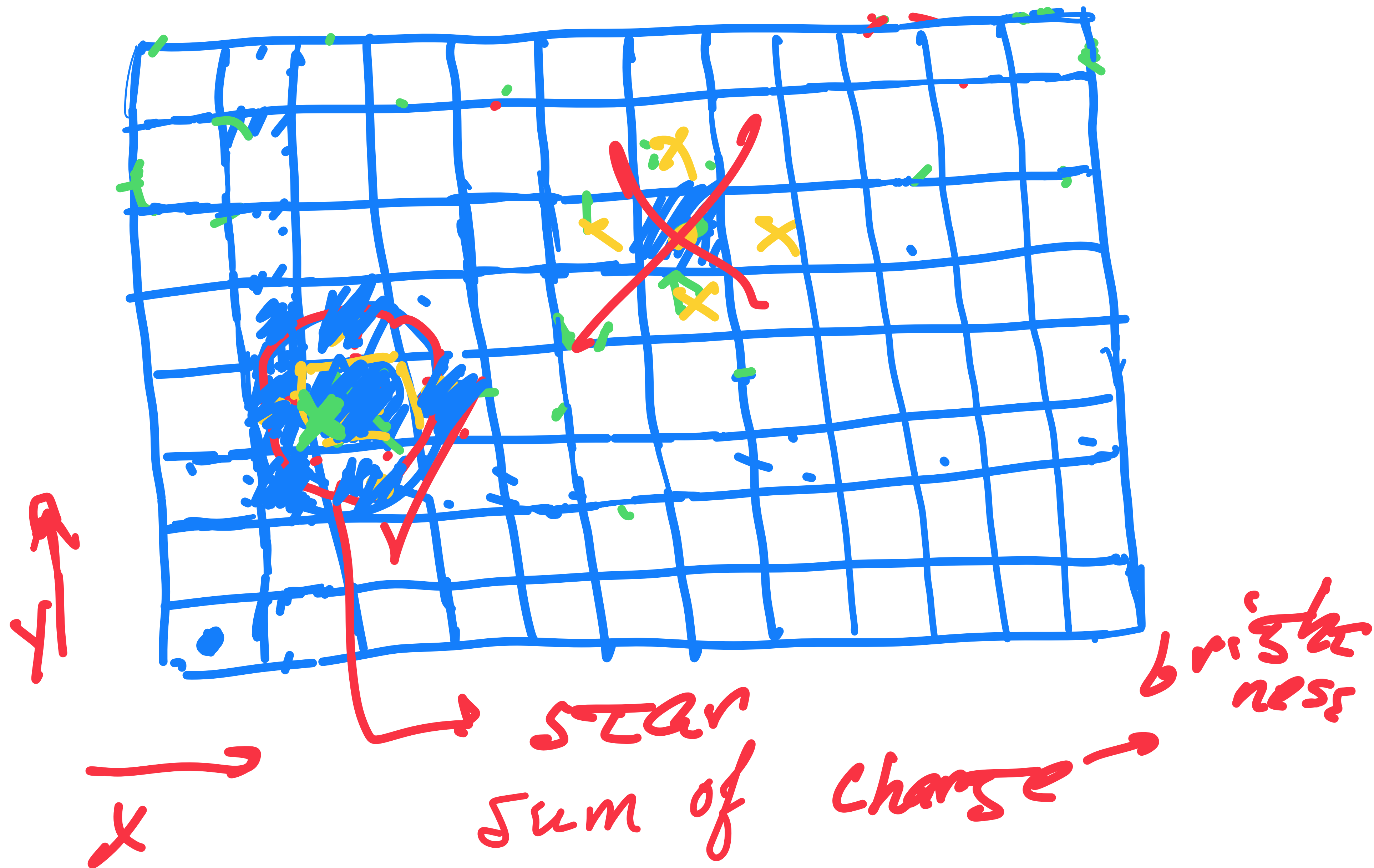
CCD READOUT

- ▶ Control electronics
 - ▶ Clock signals
 - ▶ Move charge from one pixel to next
 - ▶ Manipulation of border voltages
- ▶ Readout 1st row to readout register
 - ▶ Send to digitizing electronics
 - ▶ Move rest of image down one row
 - ▶ do it again
- ▶ Clock signals control readout



CCD PERFORMANCE: UNCORRELATED IMPACTS

- ▶ **Dark current**
 - ▶ Production of free electrons per pixel vs. time - due to thermal energy present
 - ▶ Also results noise per pixel that increases as $\sqrt{t}$
- ▶ **Malfunctioning pixels/columns**
 - ▶ **Hot pixels** - generally produces wide variations in reported charge independent of incident photons
 - ▶ **Bad columns**
 - ▶ can be a column that fails to be controlled properly
 - ▶ May be defect in pixel of readout register
 - ▶ Shows up as a column of very high charge, or no signal
- ▶ **Cosmic rays**

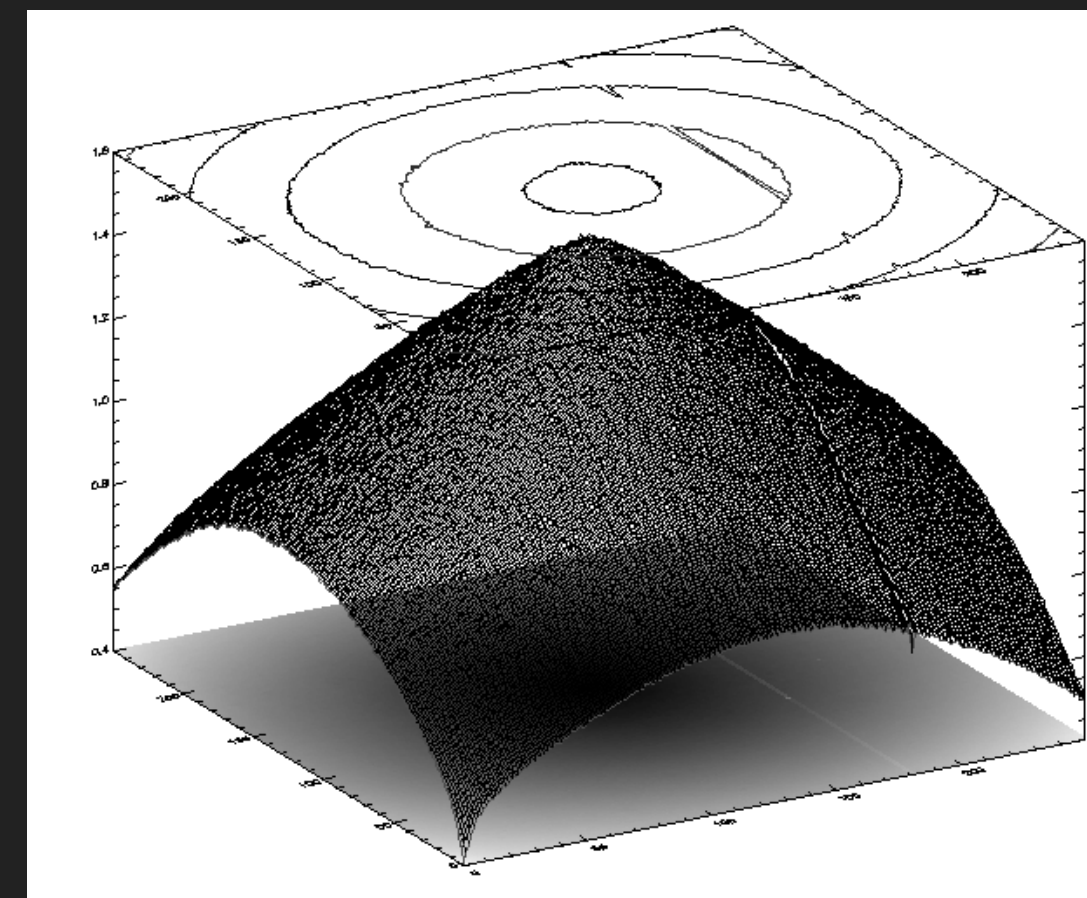


CCD PERFORMANCE: CORRELATED IMPACTS

- ▶ Several effects occur such that they scale with the # of photons incident on a pixel
 - ▶ **Pixel gain** variations
 - ▶ These are small changes in the # of electrons produced per # incident photons
 - ▶ Can be due to nonuniformities in Si, or in electronics variations controlling pixels
 - ▶ Gain variation across the field
 - ▶ **Vignetting**: Aperture of telescope blocks out light from edges of field more than center
 - ▶ Shutter sticking and other impacts on amount of light onto primary mirror
 - ▶ Obstruction from **atmospheric transparency** variations

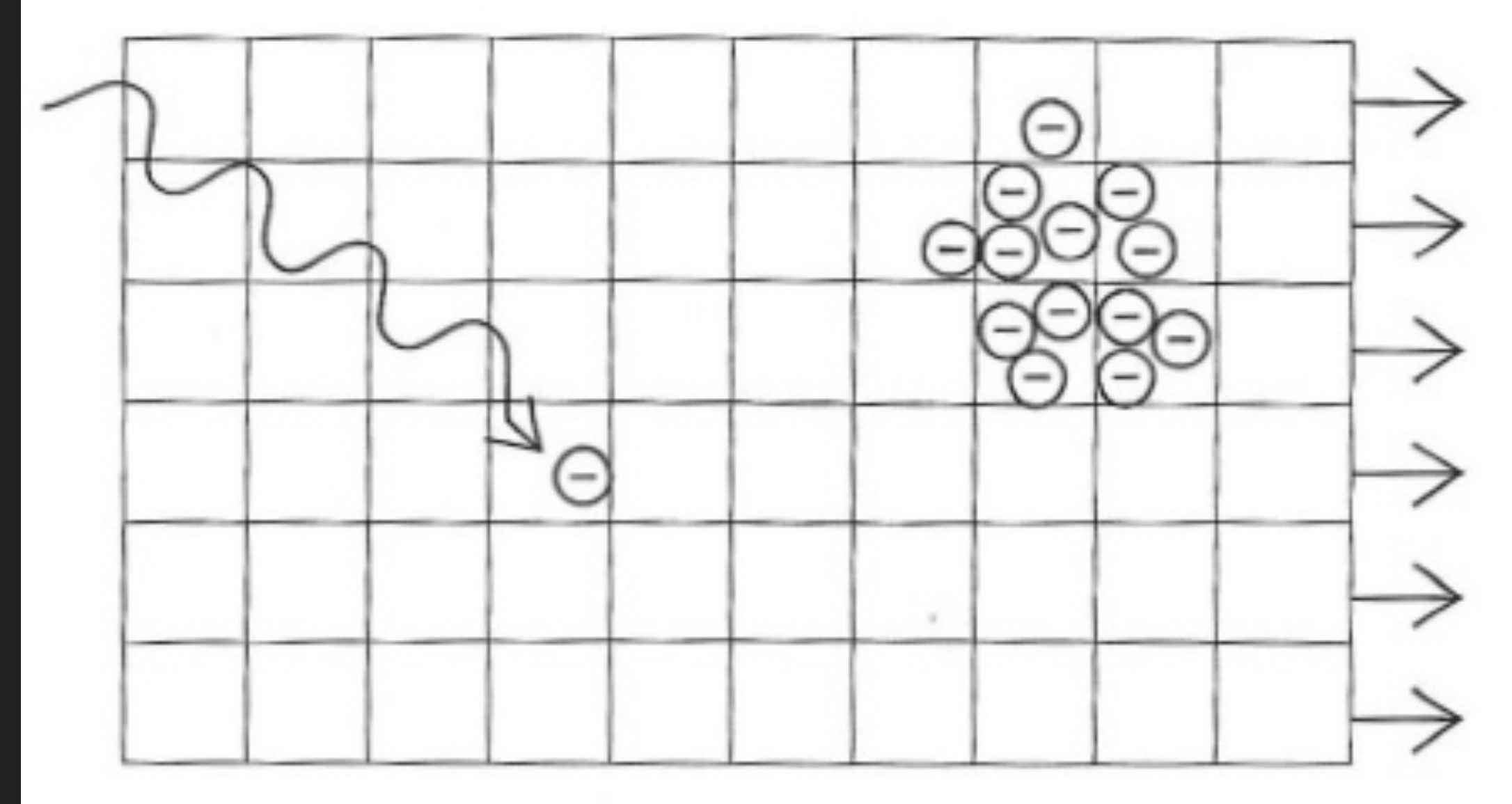
DATA REDUCTION

- ▶ Image must be corrected for correlated and uncorrelated impacts
 - ▶ Usually done with specialized images for correction
 - ▶ 'Darks': exposures with closed shutter to measure dark current and flag bad pixels
 - ▶ To correct for dark current offset per pixel
 - ▶ 'Flats': exposures with open shutter taken of a uniform illumination field, or medianing many sky fields
 - ▶ To correct for pixel gain and vignette get effects
 - ▶ Correction is then:
$$\text{Im}^{corr}(x, y) = \frac{\text{Im}^{raw}(x, y) - \text{dark}(x, y)}{\text{flat}(x, y)}$$



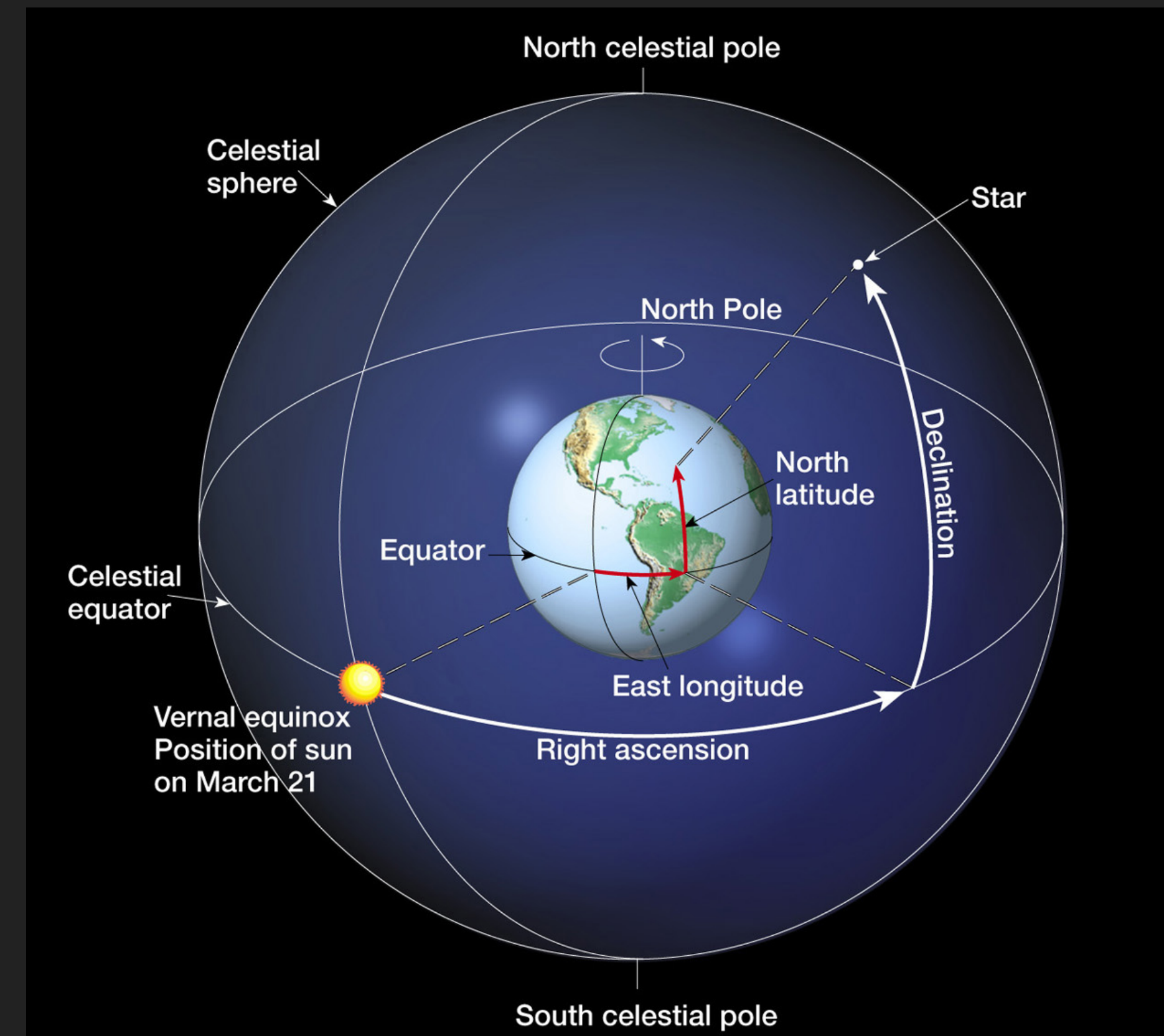
SOURCE EXTRACTION

- ▶ Once image is corrected, we can find stars
 - ▶ Apply a clustering algorithm:
 - ▶ look for patterns of close, illuminated pixels
- ▶ 'Source extraction'
 - ▶ We use Source Extractor (SEXtractor) for this
 - ▶ Some of the light coming to pixels is physical, but from the sky (eg. OH lines that deexcite)
 - ▶ Develop a model of this 'sky background' and subtract
 - ▶ Look for nearest neighbor pixels above some threshold
 - ▶ Yields
 - ▶ # counts in a 'source'
 - ▶ centroid (x,y) coordinate on the CCD



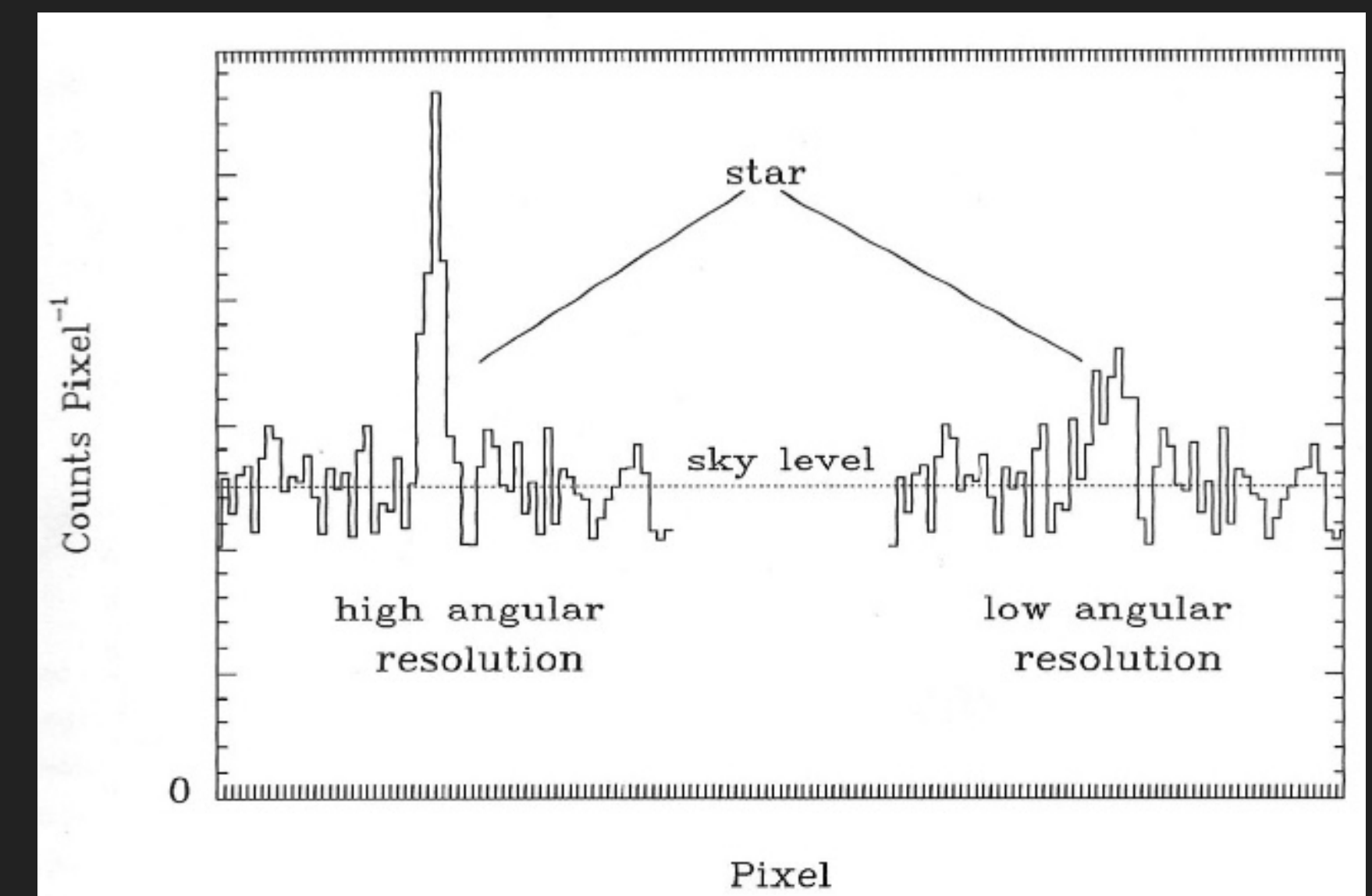
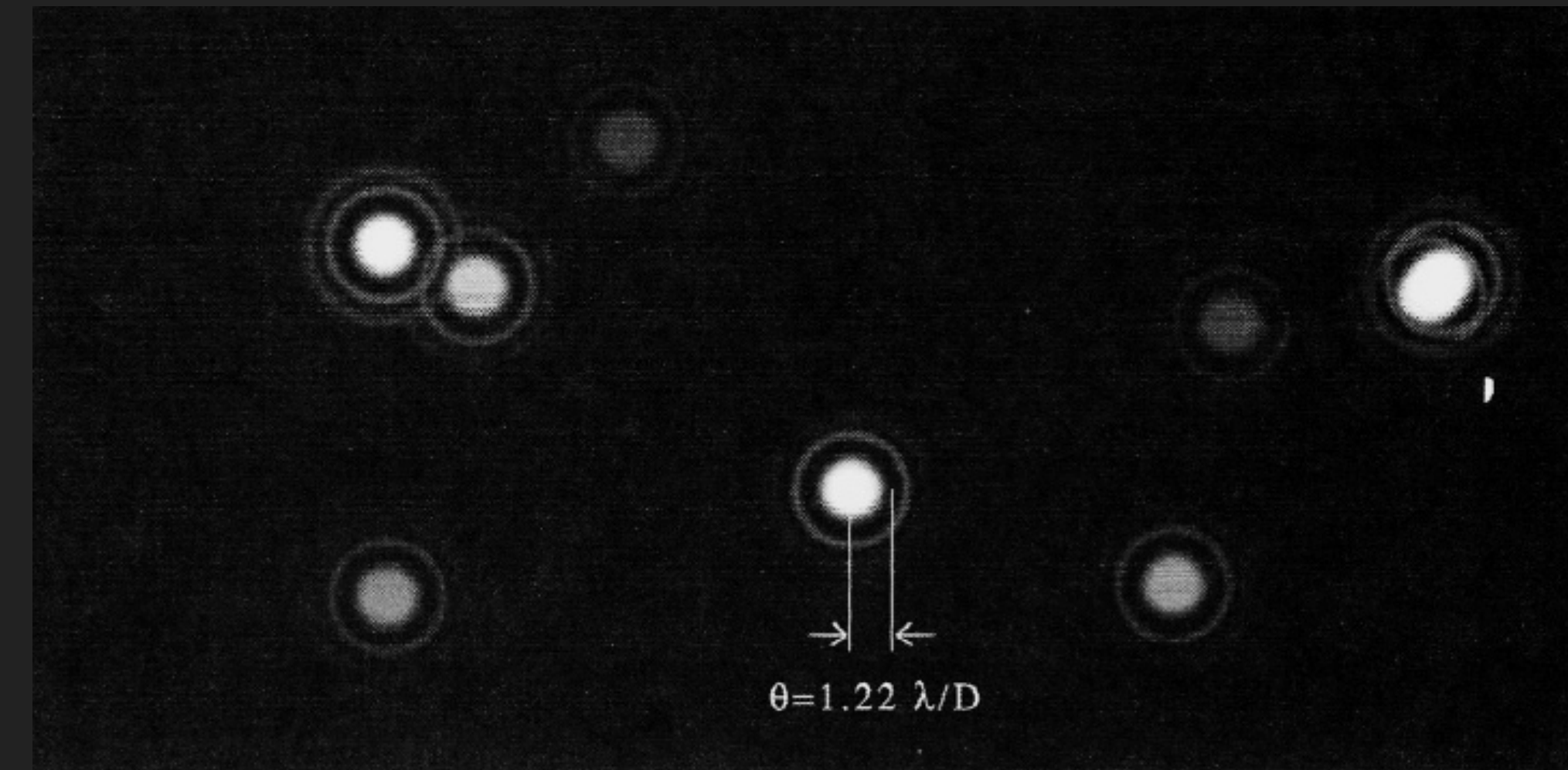
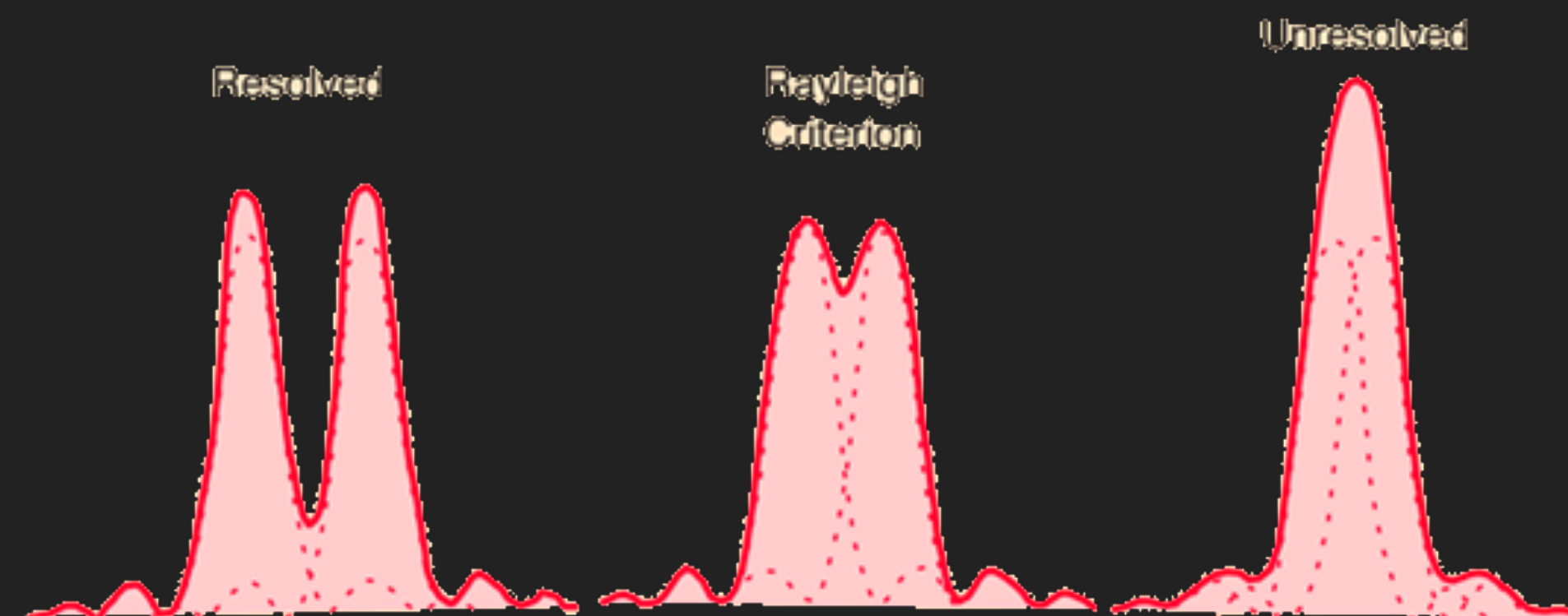
ASTROMETRY

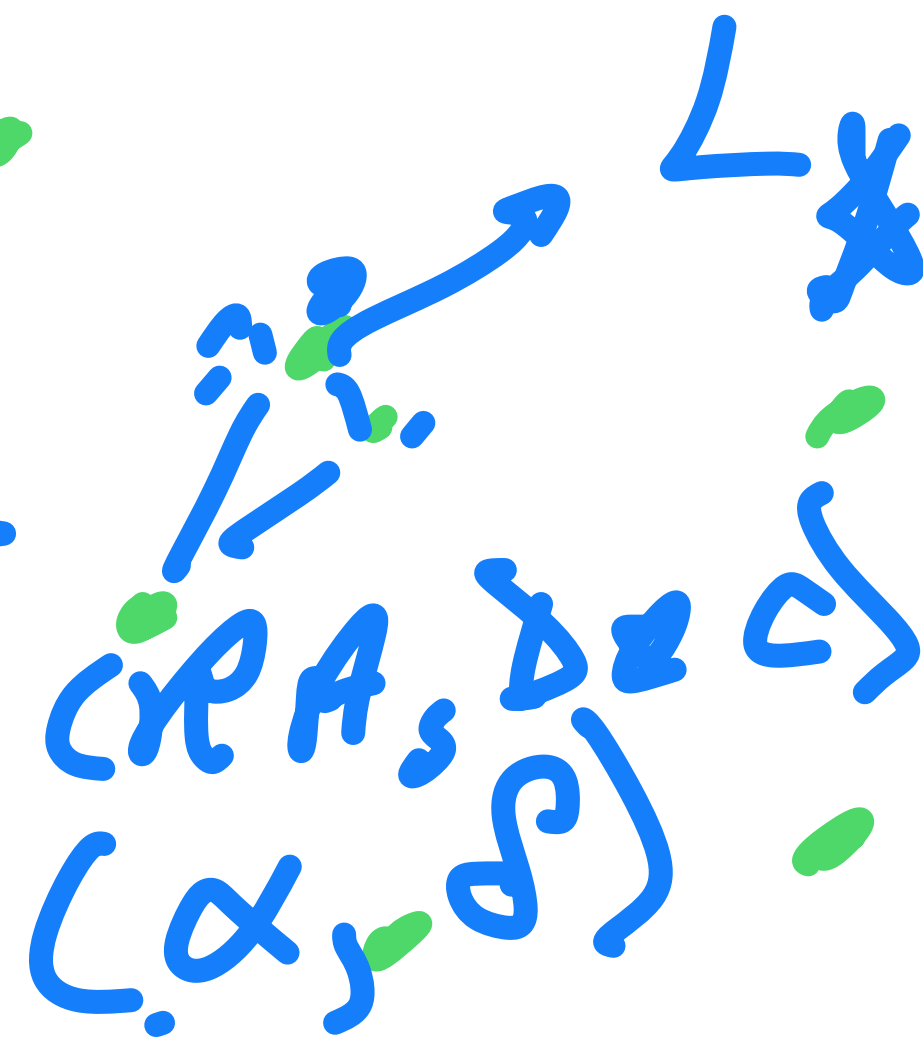
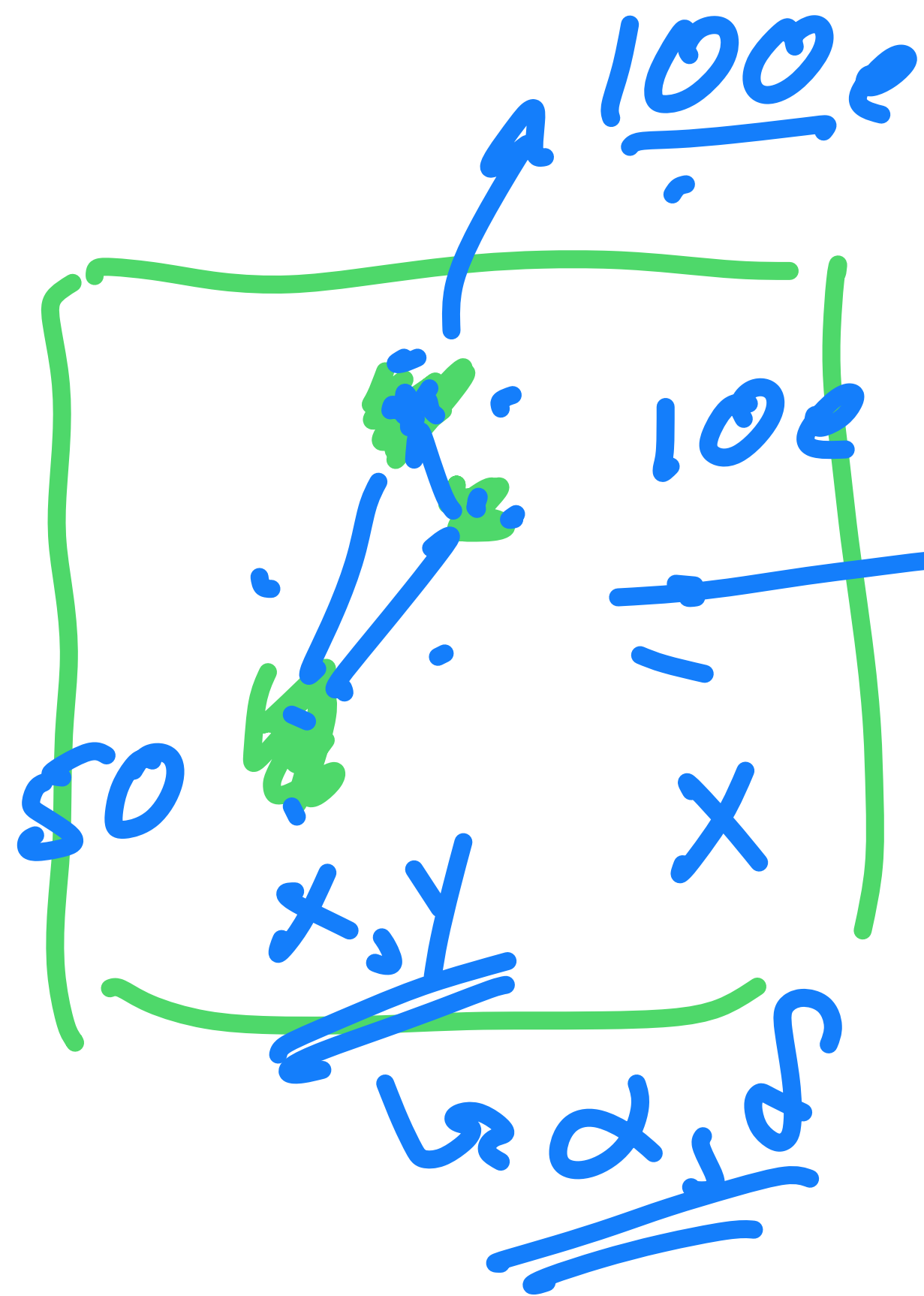
- ▶ The array of sources arranged in rectangular (x,y) CCD coordinate grid
 - ▶ This is a projection of the spherical astronomical coordinates (α, δ) onto a plane
- ▶ How to obtain astronomical coordinates
 - ▶ I.e. **Astrometry** - measure position of sources
 - ▶ Compare unique triangles of objects in (x,y) to unique triangles in a known well-measured astronomical catalog
 - ▶ Eg. USNO catalog
 - ▶ Fit to 2D analytic functions transforming (x,y) to (α, δ)
 - ▶ Termed 'matching'



ASTROMETRIC PERFORMANCE

- ▶ Several effects can spread out image of a star
 - ▶ Diffraction limit:
 - ▶ Rayleigh criterion $\sin \theta_R = 1.22 \frac{\lambda}{d}$
 - ▶ Atmospheric turbulence: 'seeing'
 - ▶ Characterize star by 'point-spread function' or PSF
 - ▶ A larger PSF
 - ▶ Integrates more noise
 - ▶ Prevents distinguishing nearby sources





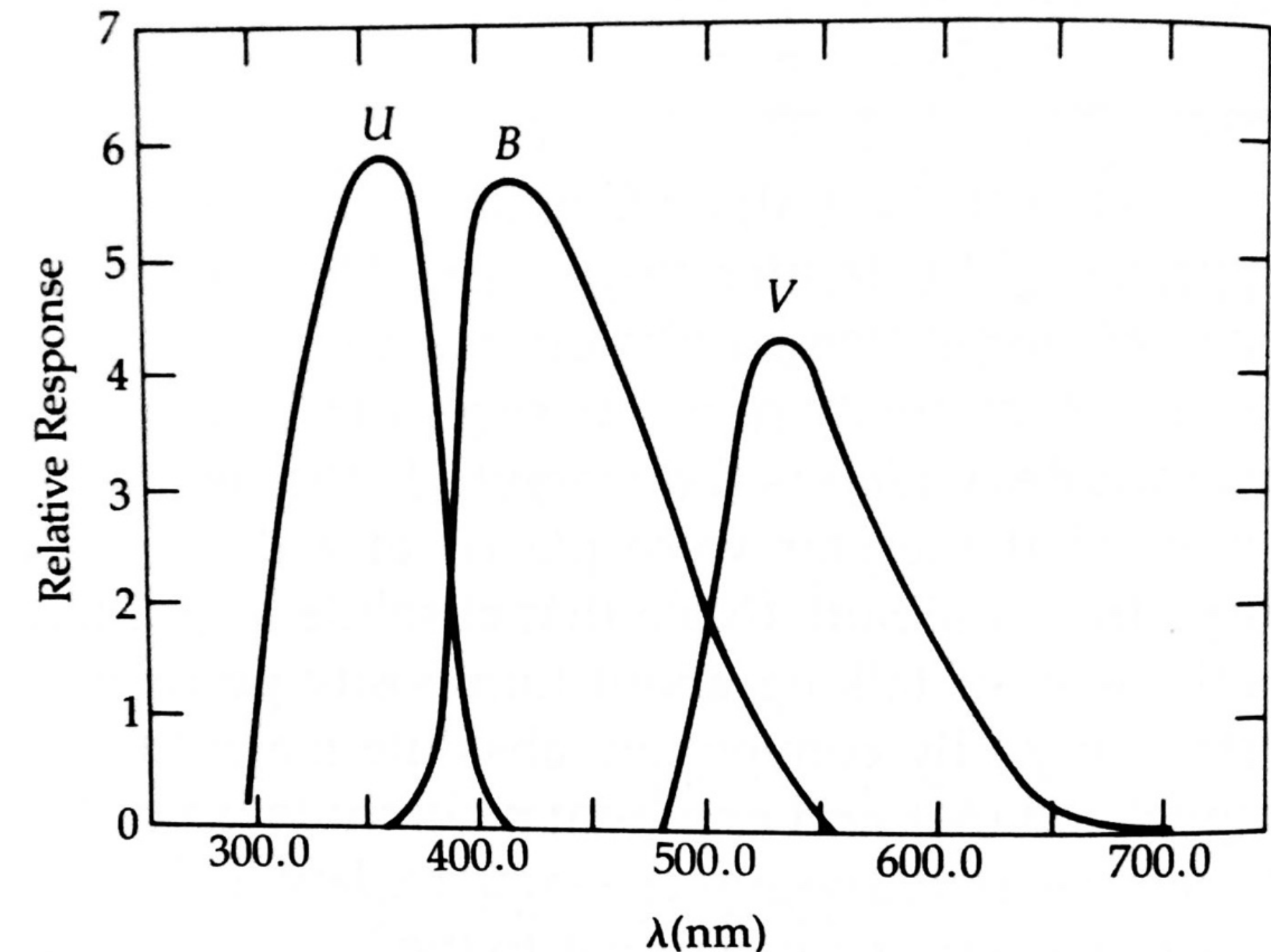
How many electrons
corresponds
to luminosity

PHOTOMETRY

- ▶ Each source has physical coordinates, but still # counts in terms of measure of brightness
 - ▶ We want brightness in some absolute measure of flux
 - ▶ Eg. Magnitudes
- ▶ Now that we have the coordinate transformation
 - ▶ We know what all of observed sources are in terms of USNO stars
 - ▶ Compare measured # counts to the magnitudes for V or R band in USNO
 - ▶ Fit the transformation relating the two
 - ▶ Also determine, for field, magnitude at which efficiency to observe goes to 50% – 'limiting magnitude' M_{LIM}
 - ▶ Yields catalog of (mag, α , δ) from (#counts, x, y)
 - ▶ While all stars are used in the comparison, including variable stars, the average transformation is dominated by nonvariables and so we can still see them

PHOTOMETRY AND LIGHTCURVES

- ▶ photometric bands
 - ▶ Narrow bands at UV, optical or infrared wavelengths
 - ▶ Obtain magnitudes in these bands
 - ▶ Several systems
 - ▶ Johnsons-Cousins: UBVRI
- ▶ Once each calibrated object list is made
 - ▶ can be easily matched to each other
 - ▶ since they are all in absolute common coordinate system (α, δ)
 - ▶ How a lightcurve assembled for all objects in a field

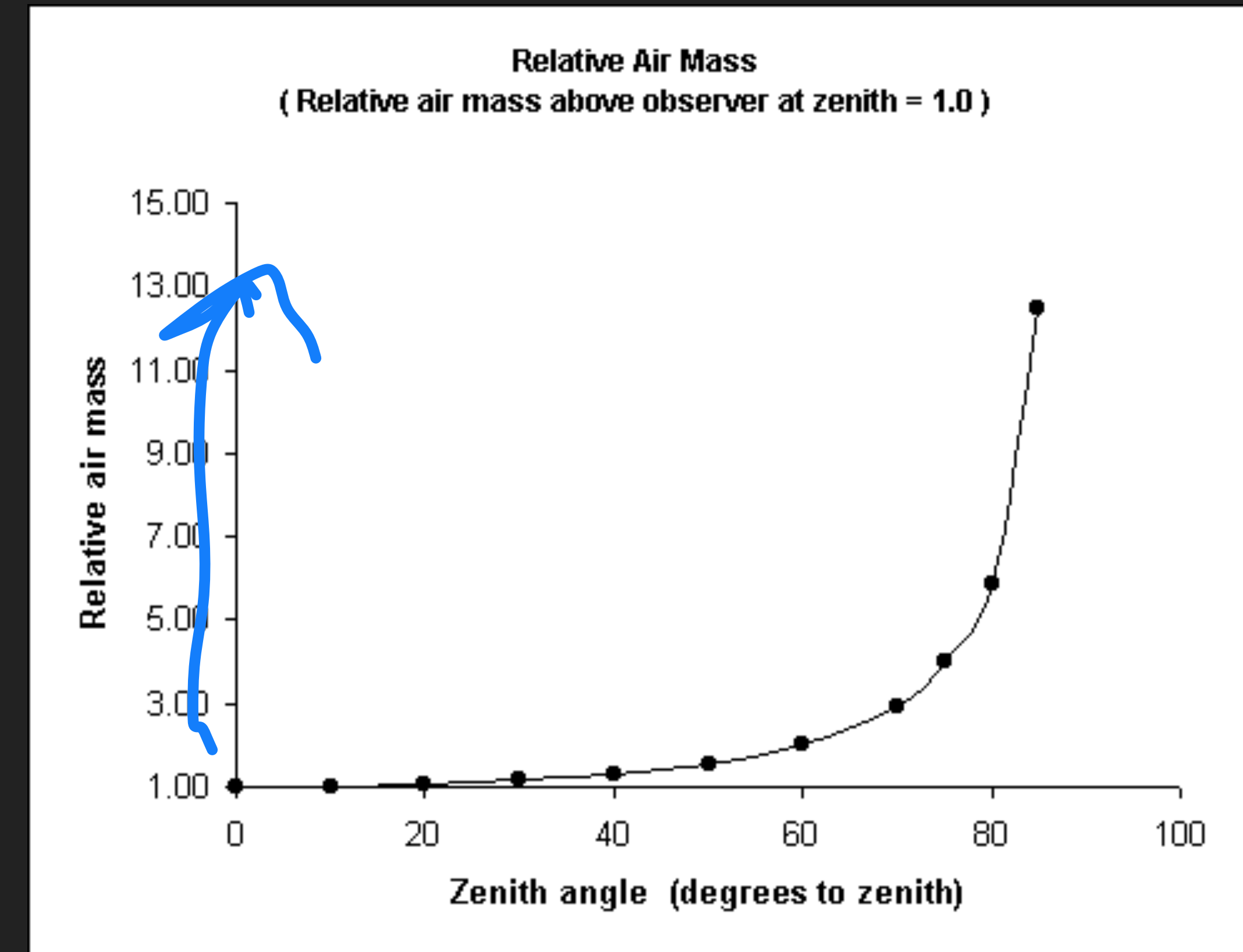


TELESCOPE OPERATIONS

- ▶ Telescopes operated according to a scheme which is organized by **target**
 - ▶ For ROTSE-I orphans data: Gamma-ray bursts (GRBs)
 - ▶ In most cases, specific targets specified before observing
 - ▶ Eg. Specific galaxies to measure in the Dark Energy Survey
- ▶ Not all observations scheduled with same **priority**
 - ▶ eg. Variable stars may be scheduled lower priority than GRBs
 - ▶ Affects when observations occur, and their frequency

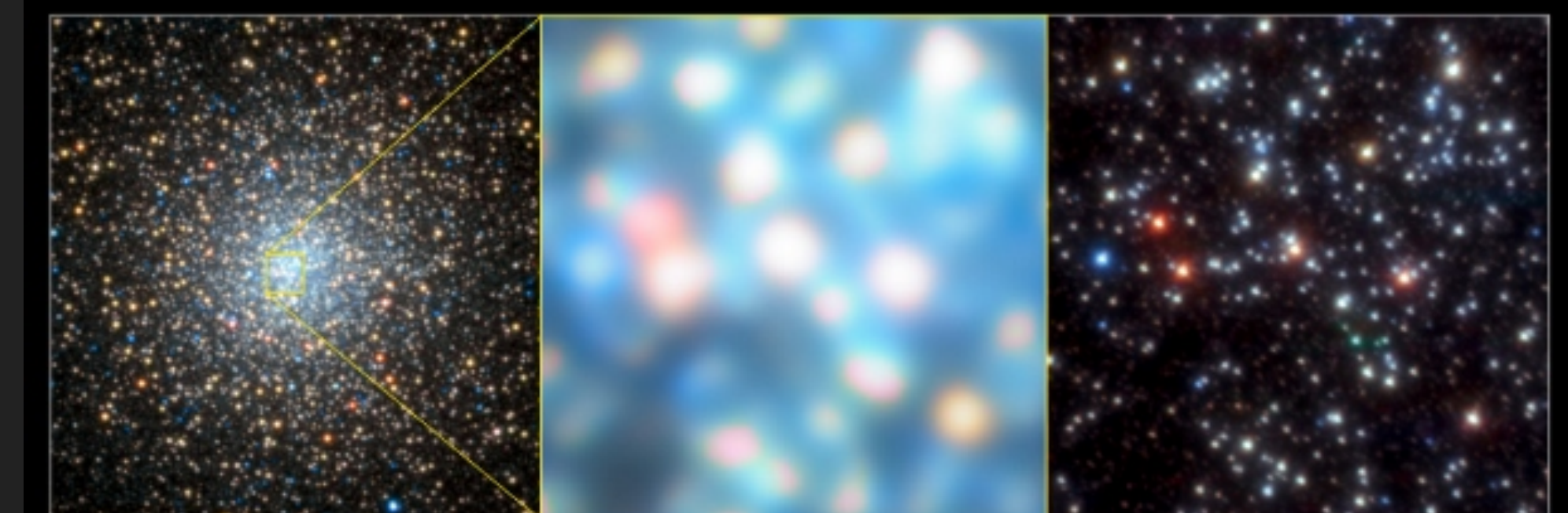
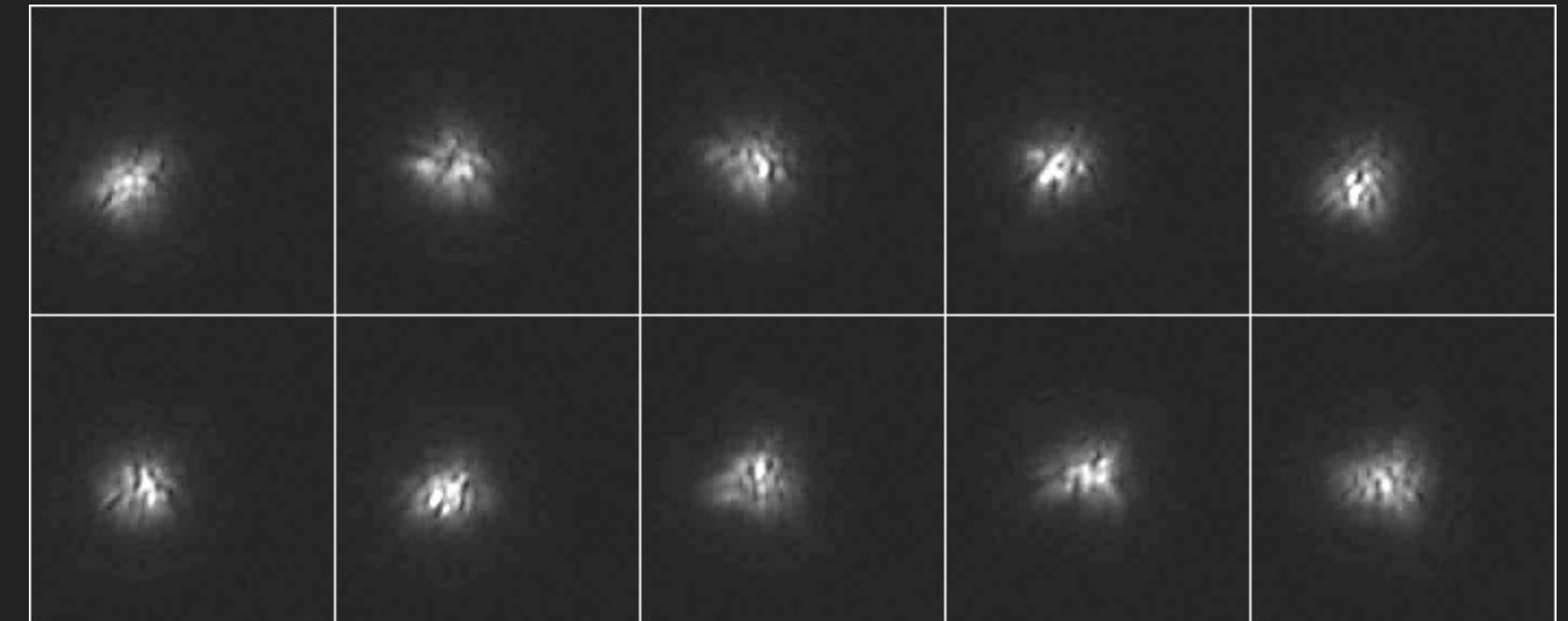
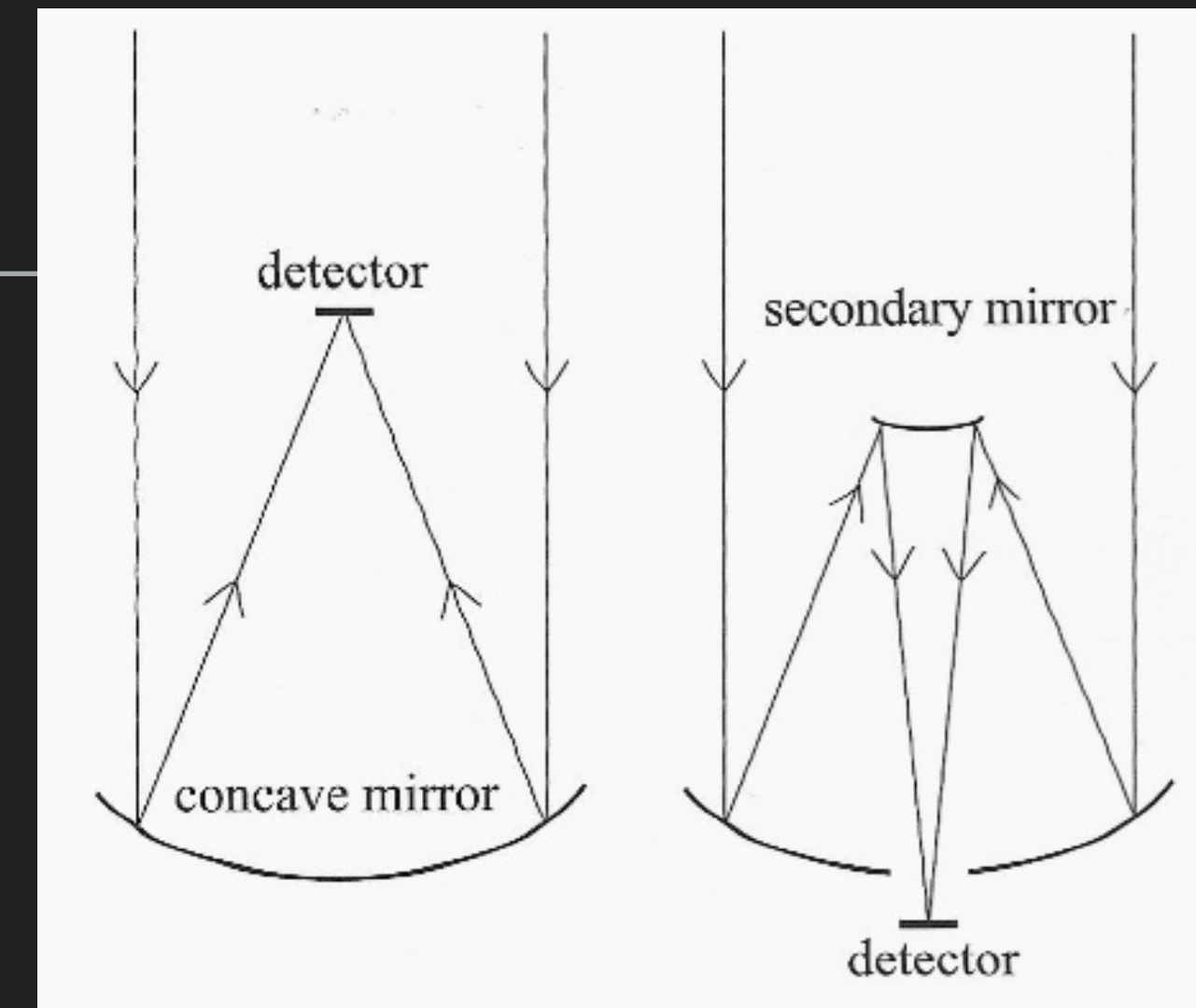
TELESCOPE OPERATIONS (CONT.)

- ▶ Daily, Seasonal and monthly cycles
 - ▶ Targets rise and set at different times depending on season
 - ▶ Moon up during observing some fraction of each month
 - ▶ Moon injects more photons into sky during observations,
 - ▶ Varies with phase
 - ▶ particularly large when near full moon
 - ▶ Sources observed lower in elevation (angle w/respect to horizon)
 - ▶ Pass thru more air (more 'airmass')
- ▶ Scheduling
 - ▶ Generally try to maximize ability to observe targets
 - ▶ Favor low airmass, low moon, away from twilight
 - ▶ Constrains when a particular field is best observed



SYSTEMATIC OBSERVING CHARACTERISTICS

- ▶ Weather presents differently at different times
 - ▶ Eg. Opaque clouds, transparent clouds
 - ▶ Sometimes limits time to observe: eg. rain, high winds
- ▶ Impacts on resolution
 - ▶ Telescope optics must be focused, and often times this can vary over time
 - ▶ Temperature shifts over night or longer can change where the focal plane lands relative to the detector, ie. Poor 'focus'
 - ▶ Atmospheric turbulence causes more smeared out stars, 'seeing'
- ▶ Even when observations can be made, these affects impact the measurement
 - ▶ Transparent clouds in a portion of a field block out some light from sources there
 - ▶ optical '**transparency**' impacts depth of field
 - ▶ Vignetting variations per exposure similar impact locally light observed from a subset of sources



VLT+MUSE Wide Field Mode
without Adaptive Optics

VLT+MUSE Narrow Field Mode
with Adaptive Optics

TELESCOPE PERFORMANCE – ASTROMETRY

► Pointing

- Telescope has a conversion between hardware coordinates (on the mechanical wheels on both axes) corresponding to RA and Dec
- Always some (small) error in reaching precise coordinates
 - In some cases can be significant
 - Sources at edges of field not observed, or observed anew

► Tracking

- Telescope must adjust pointing in real-time as Earth rotates
- Unevenness in wheels, or wear, can result in uneven tracking
 - ‘Dumbbells’ or elongated source shapes
 - Might lead to problems in source extraction

TELESCOPE PERFORMANCE – PHOTOMETRY

► Saturation

- The potential well of each pixel has finite height to its sides
 - Can only hold so many electrons before some 'spill out' – saturation
 - Electrons spill out in vertical, 'y', direction where potential barriers less (readout direction)

► Hot pixels

► Coherent noise in CCD readout

- When the dark current of adjacent/nearby rows is correlated
 - Cannot adequately subtract during data reduction, since dark current is different

► Blending

- Two nearby objects are extracted as one object
- Or their division into two objects changes each exposure as exposure conditions change
- Also can impact astrometry

